

ABSTRACT

Narayana Career Connect is a job recommendation platform built to help software engineering graduates find the right opportunities based on their skills and interests. For many fresh graduates, traditional job portals can be overwhelming and often fail to provide relevant matches, making the job search process frustrating. This platform changes that by offering a smarter, more personalized approach—focusing exclusively on fresher-friendly roles that align with individual skill sets.

But job hunting isn't just about finding openings—it's about being prepared. That's why **Narayana Career Connect** goes beyond job listings to offer mentorship and learning resources. Graduates can access expert-curated materials, industry insights, and practical guides to sharpen their skills. The platform also connects them with experienced mentors who provide career advice, technical learning, and real-world industry knowledge. This guidance helps job seekers build confidence and become job-ready.

What sets this platform apart is its strong sense of community. Graduates can interact with peers, share their job search experiences, and discuss industry trends. It's a space where users don't just find jobs—they grow together. Once they land a job, they can share their journey, inspiring and guiding others who are just starting out.

By blending **job discovery, mentorship, and community support**, Narayana Career Connect creates a seamless transition from academics to the professional world. Unlike generic job portals, it's designed specifically for fresh graduates, ensuring that every opportunity is relevant and valuable.

With its holistic approach, **Narayana Career Connect** doesn't just help graduates find jobs—it empowers them to build successful careers, make meaningful connections, and confidently step into the world of work.

KEYWORDS: Job recommendation, Fresher jobs, Skill-based matching, Career mentorship, Learning resources, Software engineering careers, Graduate job search, Professional development, Networking for freshers, Career growth

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1. INTRODUCTION

1.1 PREAMBLE FOR JOB MATCHING PLATFORM

Landing the right job as a fresh graduate isn't always easy—traditional job portals can feel overwhelming and often miss the mark in matching opportunities with your skills. **Narayana Career Connect** is here to change that by offering a personalized job matching experience tailored specifically for software engineering graduates. It helps you discover fresher-friendly roles that truly align with your expertise, interests, and career goals.

But finding a job is just one part of the journey. That's why the platform also provides mentorship, industry insights, and learning resources to help you grow. You can connect with experienced mentors, gain valuable guidance, and build confidence in your career path. With a strong community of learners and job seekers, **Narayana Career Connect** makes the transition from student life to the professional world smoother and more empowering.

1.1.1 IMPORTANCE OF NARAYANA CAREER CONNECT

Fresh graduates often face difficulties in finding jobs that align with their skills and career goals. **Narayana Career Connect** helps by providing personalized job recommendations, mentorship, and learning resources. By bridging the gap between education and employment, it empowers job seekers to start their careers with confidence.

1.1.2 PURPOSE OF THE APPLICATION

This application is designed to simplify the job search process for fresh graduates by providing a user-friendly interface for:

- **Job Recommendation:** Receive personalized job suggestions based on skills and career interests, ensuring fresher-friendly opportunities.
- **Mentorship & Guidance:** Connect with experienced mentors for career advice,

technical learning, and industry insights to enhance professional growth.

- **Learning Resources:** Access curated educational materials, industry trends, and skill-building content to stay competitive in the job market.
- **Community Engagement:** Interact with peers, share job search experiences, and exchange real-world insights to foster a strong professional network.

By integrating job discovery, mentorship, and learning, **Narayana Career Connect** empowers fresh graduates to transition seamlessly from academia to the professional world.

1.1.3 PRIVACY AND SECURITY

We are committed to protecting your personal information and ensuring the highest standards of data security. Your job preferences, mentor interactions, and learning progress will be handled with the utmost confidentiality. Our system complies with all relevant privacy regulations, ensuring a secure and trustworthy experience for all users.

1.1.4 JOIN OUR COMMUNITY

By using this platform, you're becoming part of a supportive community of fresh graduates and professionals working toward career success. We encourage you to explore opportunities, connect with mentors, and build meaningful relationships with like-minded peers. Finding a job is just the beginning—share your experiences, learn from others, and grow together in a space designed to help you thrive.

Thank you for choosing **Narayana Career Connect**. Let's make job searching easier, learning more accessible, and career growth a journey we take together!

1.2 Motivation for the Project

The motivation for developing **Narayana Career Connect** arises from the challenges fresh graduates face in finding suitable job opportunities that match their skills and career goals. Traditional job portals often fail to provide tailored recommendations for

entry-level candidates, leading to frustration and missed opportunities. This platform aims to bridge the gap between education and employment by offering personalized job recommendations, mentorship, and learning resources.

1.2.1 Addressing Job Market Challenges

Many fresh graduates struggle to find jobs that align with their skills, as employers often seek experienced candidates. This platform addresses these challenges by:

- **Skill-Based Job Matching:** Helping job seekers discover opportunities tailored to their expertise and career aspirations.
- **Fresher-Focused Opportunities:** Prioritizing entry-level job listings to ensure graduates find relevant positions quickly.

1.2.2 Enhancing Career Growth and Learning

Beyond job searching, professional growth requires mentorship and access to industry insights. This platform enhances career development by:

- **Mentorship & Guidance:** Connecting users with industry professionals who provide career advice and technical support.
- **Learning Resources:** Offering curated educational materials, industry trends, and skill-building content to help graduates stay competitive.

1.2.3 Building a Supportive Community

Networking and peer interactions play a crucial role in career success. **Narayana Career Connect** fosters a collaborative environment by:

- **Peer-to-Peer Interaction:** Encouraging graduates to share experiences, insights, and job search strategies.
- **Success Stories & Motivation:** Allowing users to document and share their career journeys, inspiring others in the community.

1.2.4 Leveraging Technology for Better Opportunities

Technology-driven platforms can streamline job searching and professional development. This project ensures accessibility and efficiency through:

- **User-Friendly Interface:** A seamless platform designed for effortless job discovery and career navigation.
- **Data Security & Privacy:** Implementing robust security measures to protect user information and interactions.

The development of **Narayana Career Connect** is driven by a deep commitment to empowering fresh graduates, ensuring they transition into the professional world with confidence. By addressing job market challenges, fostering learning, and building a strong community, this platform provides a structured and supportive pathway for career success.

1.3 SCOPE OF THE PROJECT

Narayana Career Connect aims to simplify job discovery, career growth, and mentorship for fresh graduates. The platform will feature an intuitive interface where users can register, create profiles, and showcase their skills, qualifications, and career preferences. A personalized job recommendation system will match users with fresher-friendly job opportunities based on their expertise and interests, ensuring an efficient job search experience.

Beyond job discovery, the platform will incorporate mentorship and learning resources to support professional development. Users will have access to expert-curated educational materials, industry insights, and career guidance to improve their employability. Additionally, it will foster a collaborative community where graduates can interact with mentors, share experiences, and engage in discussions on industry trends. Automated notifications will keep users informed about job openings, mentorship sessions, and career-related updates.

Security and data privacy will be a top priority, with stringent measures implemented to protect user information in compliance with relevant regulations. The system will be designed for scalability and efficiency, ensuring smooth performance as the user base expands. Regular updates and enhancements will be provided to improve functionality, user experience, and security.

This platform will serve as a one-stop destination for job seekers to enhance their skills, connect with industry professionals, and successfully transition into the workforce.

1.4 OVERVIEW OF THE PROJECT

Narayana Career Connect is a job recommendation platform designed to support fresh graduates, particularly in software engineering, by connecting them with relevant job opportunities. Unlike traditional job portals, which often overwhelm users with generic listings, this platform focuses on fresher-friendly roles, making the job search process more efficient and targeted.

Users can create and manage profiles by providing their educational background, skills, and career interests. The platform allows them to track job applications, receive notifications about new opportunities, and stay updated on their application status. Additionally, it offers a structured approach to career growth by providing mentorship, resume-building assistance, and interview preparation support.

Mentors play a crucial role in guiding students through their job search journey. They can add or remove learning resources based on student requests, ensuring that users have access to relevant and up-to-date materials. Mentors also have the ability to manage students, offering personalized advice, tracking progress, and assisting in career development.

Beyond job discovery, the platform fosters a collaborative environment where users can interact with peers, share job search experiences, and gain industry insights.

1.5 LITERATURE SURVEY

1.5.1 JOB RECOMMENDATION PLATFORM FOR FRESH GRADUATES

TITLE: Job Recommendation Platform for Fresh Graduates

AUTHORS:

- R. Karthikeyan
- P. Srinivasan
- A. Rajesh
- S. Mehta

This system is designed to bridge the gap between fresh graduates and job opportunities by providing a structured, user-friendly job search experience. Traditional job portals often lack personalized recommendations for freshers, making job hunting a challenging process. This platform addresses that gap by streamlining job discovery based on user profiles.

Users can register by providing details such as their education, skills, and career preferences. The system maintains a structured database of job openings, ensuring that users receive relevant job recommendations. Additionally, users can track job applications and receive timely notifications regarding new opportunities. The platform also includes mentor support, enabling experts to guide students, manage learning resources, and provide career advice.

Employers can post job listings targeted at fresh graduates and track applications. The platform also facilitates interactions between recruiters and candidates, making the hiring process more efficient. It serves as a centralized hub for career growth, offering resume-building guidance, interview preparation, and career development resources.

ADVANTAGES:

- **Targeted Job Recommendations:** Unlike generic job portals, the platform focuses on freshers, providing tailored job opportunities.

- **Structured Career Support:** Mentors can assist students by managing resources and offering guidance.
- **User-Friendly Interface:** The platform is designed to ensure smooth navigation and accessibility.
- **Efficient Communication:** Recruiters and job seekers can connect easily, reducing hiring time.

DISADVANTAGES:

- **Dependency on Internet Access:** Users require stable internet connectivity to access job listings and updates.
- **Profile Maintenance:** Users must keep their profiles updated to receive accurate job recommendations.
- **Limited Opportunities in Some Fields:** Fresh graduates in niche industries may find fewer job openings.

1.5.2 LINKEDIN JOB SEARCH

TITLE: LinkedIn Job Search

COPYRIGHT: LinkedIn Corporation

YEAR: 2023 (Last Updated)

METHOD:

LinkedIn Job Search is a widely used platform where job seekers create profiles, list their skills, and connect with employers. Users receive job recommendations based on their profile data, location, and industry preferences. Recruiters can search for candidates, post job openings, and directly communicate with potential hires.

DISADVANTAGES:

- **Job Recommendations are Generic:** The platform does not specifically focus on fresh graduates, leading to mismatched job suggestions.
- **Premium Features Required:** Some essential job search tools are locked behind paid subscriptions.

- **Competitive Market:** Due to high user traffic, freshers often face tough competition for job openings.

1.5.3 ONLINE JOB PORTAL MANAGEMENT SYSTEM

TITLE: Online Job Portal Management System

AUTHORS:

- A. Sharma
- M. Gupta
- P. Ramesh
- K. Patel

YEAR: 2021

METHOD: MySQL

JOURNAL: IEEE

Online job portals aim to provide a comprehensive platform for job seekers and employers. This system was developed to enhance the job search experience by providing structured job postings, application tracking, and candidate management. The centralized database stores job listings and applicant data, enabling employers to filter and shortlist candidates efficiently.

Users can search for jobs based on skills, location, and industry, while recruiters can manage job postings and track applications. The system ensures that job seekers receive real-time updates on job applications and interview schedules.

ADVANTAGES:

- **Efficient Job Search:** Job seekers can filter jobs based on their qualifications.
- **User-Friendly Interface:** Simple navigation for both job seekers and recruiters.
- **Application Tracking:** Candidates can monitor their application status in real time.

- **Resume Storage:** Users can upload and manage multiple resumes.

DISADVANTAGES:

- **Internet Dependency:** Users must have an active internet connection to access job listings.
- **Data Overload:** Large job databases can overwhelm users with excessive options.
- **Profile Verification Required:** Employers often require strict verification before considering applications.

1.6 PROBLEM STATEMENT

In today's competitive job market, fresh graduates often face significant challenges in securing suitable employment opportunities. Many lack the necessary industry connections, relevant work experience, and guidance needed to navigate their career paths effectively. Traditional job portals cater to a broad audience but fail to provide targeted opportunities and personalized career support for freshers. As a result, job seekers struggle with mismatched job recommendations, lack of visibility to recruiters, and inefficient application tracking.

Additionally, graduates often find it difficult to access career resources such as mentorship which are crucial for securing their first job. The absence of structured support systems results in missed opportunities and prolonged job search periods. Recruiters, on the other hand, face difficulties in identifying fresh talent that matches their requirements due to the lack of a centralized platform dedicated to fresh graduates.

To address these challenges, our project aims to develop **Narayana Career Connect**, a comprehensive job recommendation and career development platform specifically designed for fresh graduates. The platform will streamline the job search process, facilitate seamless interactions between job seekers and recruiters, and provide essential career resources. By offering structured job recommendations, mentorship opportunities, and an intuitive application tracking system, **Narayana Career Connect** will help

bridge the gap between graduates and employment opportunities, ensuring a smoother transition from education to the workforce.

1.7 OBJECTIVES OF THE PROJECT

Skill-Based Job Matching: The primary objective is to enable students to manage their skills by adding programming languages and other competencies such as Python, C, Java, etc. Based on these skills, the system will list real-time fresher job openings from professional job platforms like LinkedIn, Glassdoor, and others, ensuring students have access to the latest employment opportunities in their relevant fields.

Mentorship and Guidance: Provide students with the ability to request mentorship for career development, access to learning resources, and professional suggestions. By interacting with experienced mentors, students can gain insights into industry trends, receive personalized guidance, and make informed career decisions.

Networking and Collaboration: Foster a professional network where students can connect with peers who have similar career interests. This will encourage collaboration, knowledge sharing, and discussions on career growth, technical skills, and job opportunities.

1. Facilitate Student Profile and Skill Management

- Develop an intuitive dashboard where students can add, manage, and update their skills.
- Implement a structured skill categorization system to help students showcase their expertise efficiently.
- Ensure seamless navigation and user-friendly interaction with the profile management system.
- Allow students to track their progress in terms of skill development and career growth.

2. Real-Time Job Listings Based on Skills

- Fetch and display real-time job openings from trusted job platforms like LinkedIn , Glassdoor and all other job search platform.
- Categorize jobs based on skill sets to ensure students find relevant opportunities quickly.
- Provide filters to refine job searches based on location, industry, and company preferences.
- Ensure continuous updates of job listings to maintain relevance and accuracy.

3. Enable Mentorship Requests and Professional Interaction

- Allow students to request specific resources from mentors based on their career goals.
- Facilitate direct interaction between students and mentors for professional guidance.
- Maintain a mentorship tracking system to help students follow up on their progress.
- Enable mentors to provide curated resources, training materials, and career advice.

4. Student Networking and Community Building

- Develop a platform where students can connect with peers based on similar career interests.
- Provide discussion forums and knowledge-sharing spaces for career growth and learning.
- Implement features like messaging and group collaborations to enhance professional networking.
- Encourage interactive engagement through peer-to-peer learning opportunities.

2. SYSTEM ANALYSIS

2.1. INTRODUCTION

The process of job searching and career development has become increasingly complex, with numerous platforms providing scattered opportunities that are difficult for students to track effectively. Many fresh graduates struggle to find jobs that align with their skills due to a lack of personalized job recommendations and professional guidance. Additionally, there is a gap in mentorship and networking opportunities that could help students transition smoothly into their careers.

To address this challenge, a comprehensive solution has been developed in the form of the **Narayana Career Connect** application, designed to streamline the job searching and career development process for students. The Narayana Career Connect platform features an integrated framework that leverages real-time job listings from multiple professional platforms such as LinkedIn, Glassdoor, and Indeed. It also includes an advanced career recommendation system that analyzes student profiles, skills, and interests to match them with suitable job openings.

The Narayana Career Connect application incorporates a structured student profile system where users can add and manage their skills, certifications, and career interests. This information is securely stored in a centralized database, allowing the platform to provide tailored job recommendations based on real-time industry demands. The system also enables students to request mentorship, access learning resources, and engage in discussions with professionals to receive career advice.

To enhance user engagement, Narayana Career Connect includes a **networking feature** that allows students to connect with peers who share similar career goals. The platform provides discussion forums and direct messaging functionalities, enabling collaboration and knowledge exchange. These features aim to bridge the gap between education and employment by fostering a professional community where students can interact, seek guidance, and develop industry-relevant skills.

Furthermore, Narayana Career Connect ensures **real-time synchronization** with job databases, keeping students informed about the latest opportunities that match their skill

sets. The platform prioritizes data accuracy and user accessibility, ensuring that students receive up-to-date career guidance, job alerts, and mentorship recommendations.

By integrating job listings, mentorship, and networking into a single platform, Narayana Career Connect empowers students to take control of their career development, enhancing their employability and connecting them with real-time job opportunities in an efficient and structured manner.

2.2. SYSTEM STUDY

The system study phase of the Narayana Career Connect application is a critical step aimed at thoroughly understanding the job searching and career development processes, user requirements, and technical considerations necessary for developing an effective software solution. This phase begins with an exhaustive examination of current practices within the career development ecosystem, including student registration, skill management, and job matching procedures. Through meticulous observations and stakeholder consultations, user requirements are meticulously gathered to understand the needs and preferences of students, mentors, and recruiters. This involves understanding preferences for skill-based job recommendations, mentor interactions, and networking opportunities.

Concurrently, a comprehensive analysis of technical requirements ensures alignment with critical considerations such as data security, scalability, and interoperability with existing job platforms. This includes assessing compatibility with job listing sources like LinkedIn and Glassdoor, integration capabilities with career mentorship systems, and compliance with regulatory guidelines governing data privacy. The system study phase also involves identifying pain points and areas for improvement within the job searching and networking process, such as inefficiencies in skill evaluation or challenges in connecting with mentors. By documenting these findings, the objectives and scope of the Narayana Career Connect application are clearly delineated, outlining the functionalities and features necessary to address identified needs and challenges effectively.

Ultimately, the system study serves as the foundation for the successful development of the Narayana Career Connect application, providing a roadmap for the project team to create a software solution that streamlines skill-based job searching, enhances communication and coordination among stakeholders, and ultimately helps students build successful careers.

- **Login:** It is used to authenticate the user with email and password.
- **Registration:** It is used to provide access for new users.
- **Profile Completion:** After registration, students must fill in their profile details.
- **Manage Skills:** Students can add their skills such as Python, C, etc.
- **Job Matching:** Suggests real-time job openings based on students' listed skills from platforms like LinkedIn, Glassdoor and all other job search platforms.
- **Mentor Interaction:** Allows students to request mentorship and interact with mentors for professional guidance.
- **Networking:** Enables students to connect with other students and professionals to expand their career opportunities.

2.3. EXISTING SYSTEM

Recognizing the need for an automated system to streamline job searching and career development for freshers, existing platforms provide solutions that assist job seekers in finding relevant job openings. Several online job portals like LinkedIn and Glassdoor have been developed to address these challenges by offering job listings based on user profiles. These platforms typically utilize integrated frameworks that include extensive databases to store job postings, user profiles, and recruiter information.

At the core of these solutions is an integrated framework that includes sophisticated databases to store and manage job-related data. These databases serve as centralized repositories for storing user profiles, job applications, and employer details, allowing seamless access and retrieval of crucial information. Additionally, they facilitate analytical processing, empowering job seekers with personalized job recommendations based on their skills and preferences. By analyzing data such as industry demand, job market trends, and skill requirements, these platforms enable proactive job searching

and career planning, ensuring that freshers can connect with the right opportunities effectively.

A key feature of existing job portals is their user-friendly registration and profile creation process. Individuals can create accounts, upload their details, and specify their skills and interests. By maintaining up-to-date profiles, job seekers enhance their visibility and increase their chances of being matched with suitable job openings. This structured approach ensures that candidates are presented with relevant opportunities, aligning with their career goals and skill sets.

Moreover, these platforms incorporate job search functionalities that allow users to filter and find job opportunities based on various criteria such as location, industry, and required skills. This feature enhances job seekers' efficiency by providing them with tailored recommendations, reducing the time spent searching for suitable positions. However, despite these advancements, there remains a need for a more streamlined and student-focused system that bridges the gap between fresh graduates and recruiters, providing mentorship and skill-based job matching in real-time.

2.3.1. DISADVANTAGES

- **Complex and Non-Intuitive Interface:** Many career platforms are designed for experienced professionals, making them overwhelming for fresh graduates. The lack of a user-friendly experience discourages first-time job seekers from effectively utilizing available resources.
- **Limited Personalized Career Recommendations:** Most platforms rely on keyword-based searches rather than understanding an individual's skills, interests, and aspirations. This leads to irrelevant job recommendations, making it harder for users to find suitable opportunities.
- **Inefficient Skill-Centric Filtering:** Current job platforms prioritize past work experience over actual skills, making it difficult for fresh graduates with strong skill sets but no formal work experience to find relevant job opportunities.
- **Lack of Integrated Mentorship and Career Support:** Existing platforms often do not offer structured mentorship or career guidance, leaving freshers without proper direction regarding industry expectations, career growth paths, and skill development.

- **Delayed Job and Internship Notifications:** Many job portals fail to provide real-time updates on relevant job or internship postings, causing freshers to miss timely opportunities.
- **Scalability and Accessibility Concerns:** Some platforms struggle to accommodate a high volume of users, leading to slow website performance, login issues, and inconsistent experiences for job seekers.
- **Minimal Employer Engagement with Freshers:** Most employers on job portals primarily focus on experienced candidates, making it difficult for fresh graduates to attract recruiters' attention and secure opportunities.

2.4. PROPOSED SYSTEM

The **Narayana Career Connect** platform is designed to revolutionize the job search experience for fresh graduates by integrating AI-driven career matchmaking, real-time job alerts, and mentorship opportunities. By addressing key challenges in existing job portals, this system enhances accessibility, engagement, and efficiency in job placement for freshers.

Key Features:

1. AI-Powered Job Matching

- Matches job seekers with relevant opportunities based on skills, interests, and career aspirations rather than just keyword searches.
- Uses an intelligent ranking algorithm to suggest the best-fit job roles, internships, and training programs.

2. Real-Time Notifications

- **Instant Job Alerts:** Sends real-time notifications to users about new job openings, ensuring they don't miss out on opportunities.
- **Mentorship & Career Events:** Notifies users about career guidance sessions, webinars, and industry networking events.

3. Skill-Based Filtering & Personalized Recommendations

- Enables filtering job opportunities based on skill sets rather than just work experience, helping freshers find suitable roles.
- Suggests skill enhancement resources based on industry trends and job requirements.

4. Integrated Mentorship & Career Guidance

- Connects students with industry professionals for mentorship and career counseling.

- Offers structured guidance on industry expectations, resume building, and interview preparation.

5. Employer-Fresher Engagement Portal

- Encourages recruiters to focus on fresh talent by providing employer access to a dedicated fresher hiring database.
- Facilitates direct communication between recruiters and job seekers.

6. Scalable & Accessible Platform

- Designed to handle high traffic efficiently, ensuring a seamless user experience.
- Optimized for both web and mobile access, making job searching more convenient for users.

2.4.1. ADVANTAGES

1. Smart Powered Job Matching:

- Uses **intelligent algorithms** to match candidates with jobs based on skills, interests, and career goals.
- Reduces dependency on **keyword-based** searches, ensuring more relevant job recommendations.

2. Skill-Based Filtering:

- Focuses on **skills rather than past experience**, helping fresh graduates find suitable job opportunities.
- Suggests **career resources** and learning programs to bridge skill gaps.

3. Interactive Mentorship & Career Guidance:

- Provides access to **mentorship programs, industry experts, and career counselors** to help users make informed career decisions.
- Offers **personalized career pathways** based on user profiles.

4. Company & Role Insights:

- Offers insights into **company culture, salary trends, job growth, and required skills**.
- Provides **detailed job role descriptions** to help candidates make informed decisions.

5. Employer-Fresher Engagement:

- Encourages recruiters to **prioritize fresh graduates** by offering a **dedicated fresher hiring section**.
- Enables **direct interaction** between employers and candidates for networking opportunities.

6. Scalable & User-Friendly UI:

- Designed with a **modern, intuitive interface** to ensure smooth navigation for all users.
- **Supports future expansion**, accommodating more job postings and career resources over time.

2.5. PROPOSED METHODOLOGY

The Narayana Career Connect platform is designed to streamline **job searching, mentorship, and skill-based networking** for freshers. The system matches students with **real-time job openings** based on their **skills** and allows them to **request mentorship and career resources** while fostering a **professional student network**.

The platform consists of three **main user roles: Students (Job Seekers), Mentors, and Admins**.

1. Student (Job Seeker) Module

Students **create profiles**, add skills, and receive **job recommendations** from real-time job listings on platforms like **LinkedIn, Glassdoor, and Indeed**.

Key Features:**➤ Profile Creation & Skill Management:**

- Students enter **personal details, education, and experience**.
- Add **skills (Python, C, Data Science, etc.)** to their profile.
- AI analyzes skillset and **suggests in-demand skills** for career growth.

➤ Real-Time Job Listings & Applications:

- The system fetches **live job openings** from LinkedIn, Glassdoor, and other platforms.
- Jobs are **filtered and ranked** based on the student's skillset.
- One-click **job application tracking** within the dashboard.

➤ Mentorship & Resource Request:

- Students can **request mentorship** based on **career aspirations and skills**.
- Mentors can provide **study materials, career advice, and resume reviews**.
- **Live Q&A sessions** and webinars from industry professionals.

➤ Professional Networking:

- Students can **connect with peers**, form discussion groups, and **collaborate** on projects.
- Peer-to-peer **resume feedback and interview preparation**.

2. Mentor Module:

Mentors are industry experts or senior professionals who **guide freshers** and provide **career assistance**.

Key Features:

➤ **Profile Setup & Availability Management:**

- Mentors sign up, add expertise (e.g., Software Development, Data Science).
- Set availability for **one-on-one mentorship sessions**.

➤ **Mentorship Requests & Resource Sharing:**

- Receive mentorship requests from students based on **career interests**.
- Share **study materials, job market insights, and interview tips**.

➤ **Career Counseling:**

- Host **career-focused webinars and workshops**.
- Participate in one-to-one interaction **sessions** to answer student queries.

3. Admin Module (Management & Moderation)

Admins **oversee** platform activities, ensure data security, and manage **user interactions**.

Key Features:

➤ **User Verification & Role Management:**

- Verify **student, employer, and mentor registrations**.
- **Prevent spam/fraudulent job postings** and ensure platform authenticity.

➤ **Job Aggregation & Data Management:**

- Maintain **real-time job listing integrations** with LinkedIn, Glassdoor, etc.
- Monitor **job recommendations** for accuracy.

➤ **Monitoring & Dispute Resolution:**

- Handle **user complaints and disputes** between students and mentors.
- Ensure compliance with **platform policies and community guidelines**.

➤ **Content Moderation & Security:**

- Prevent inappropriate or **misleading job listings**.
- Ensure **secure student-mentor interactions**.

2.6. Feasibility Study

1. Technical Feasibility

The system is built using **modern, scalable, and widely supported technologies**, ensuring **high performance, security, and ease of maintenance**.

✓ **Frontend: React.js** with **Tailwind CSS** for a **responsive, fast, and modern UI**.

✓ **Backend: Node.js** with **Express.js**, offering **high scalability and efficient API handling**.

✓ **Database: MySQL**, providing **structured, relational data storage** with optimized queries for job searches and user interactions.

✓ **Integration: Fetching real-time job listings** from **LinkedIn, Glassdoor, and other platforms** using APIs.

✓ **Scalability & Security:**

- Supports **thousands of users simultaneously**.
- Uses **JWT-based authentication** for secure logins.
- **Optimized API calls** for real-time data retrieval.

Overall, the **tech stack ensures smooth functionality, scalability, and security** for students, mentors, and employers.

2. Economic Feasibility

A cost-benefit analysis confirms that **the platform is economically viable with minimal recurring costs.**

✓ **Development Costs:**

- **One-time expenses** for initial development and database setup.
- Open-source technologies (**React.js, Node.js, MySQL**) reduce licensing costs.

✓ **Operational Costs:**

- **Database hosting** (MySQL on AWS/GCP).
- **Server maintenance** (Node.js Express backend).
- **APIs for job listings** (LinkedIn, Glassdoor, etc.).

✓ **Revenue & Sustainability:**

- Potential **collaborations with companies** for direct hiring.
- **Subscription model** for premium mentorship features.
- Monetization via **job postings and career resource purchases.**

Overall, **the platform's benefits outweigh costs**, making it a **sustainable career solution.**

3. Operational Feasibility

The platform is designed to **efficiently connect students, mentors, and employers**, ensuring **high usability and acceptance.**

✓ **User-Centric Design:**

- **Easy onboarding** for students, mentors, and employers.
- **Skill-based job matching** for **real-time job openings.**
- **Mentorship request system** for **career guidance.**

✓ Job & Mentorship Coordination:

- **Automated job recommendations** based on skills.
- **One-click mentor requests** for career growth.

✓ Scalability & Future Enhancements:

- **Cloud-based deployment** ensures smooth performance.
- Future **career suggestions** based on market trends.

2.7. SOFTWARE ENVIRONMENT

The desired features and functionalities of the application from the perspective of different users.

Students:

- Register and create a profile by adding skills such as Python, C, Java, etc.
- View job listings based on skills from real-time sources like LinkedIn and Glassdoor.
- Request mentorship and access career resources from mentors.
- Interact with mentors for career guidance and professional networking.
- Connect with other students for skill-building and collaboration.

Mentors:

- Accept mentorship requests from students.
- Provide professional advice and resources related to career development.
- Share industry insights, learning materials, and interview tips.
- Engage with students through discussions and networking features.

Admin:

- Manage user accounts, verify student profiles, and approve mentors.
- Oversee job postings and ensure legitimate opportunities are listed.
- Monitor mentorship requests and student-mentor interactions.

- Generate reports on platform usage, job placements, and mentorship activities.

2.7.1 FUNCTIONAL REQUIREMENTS

User Registration and Profile Management:

- Allow users (students, mentors, employers, and admin) to register and create profiles.
- Enable students to add and update their skills dynamically.
- Verify mentors and employers before granting them access to their respective functionalities.

Job Matching System:

- Fetch real-time job openings from platforms like LinkedIn and Glassdoor based on student skills.
- Provide students with personalized job recommendations.
- Allow employers to post job opportunities directly on the platform.

Mentorship Request and Resource Sharing:

- Enable students to request mentors for career advice and required resources.
- Allow mentors to accept or decline mentorship requests.
- Provide chat or discussion features for mentor-student interactions.

Search and Filtering System:

- Implement search functionality to find jobs, mentors, and other students based on criteria such as skills, experience, and industry.
- Provide filtering options for better job and mentorship matches.

Networking and Collaboration:

- Allow students to connect with peers, share knowledge, and collaborate on projects.

- Enable students to form study or learning groups based on mutual interests.

Admin Panel for Platform Management:

- Verify and approve student profiles, mentors, and employers.
- Manage job postings to prevent spam or irrelevant listings.
- Monitor user interactions and take action on reported issues.
- Generate analytics and reports on job placements and user engagement.

2.7.2 NON-FUNCTIONAL REQUIREMENTS**Performance:**

- Ensure quick response times when fetching job listings and mentorship data.
- Optimize API calls to handle multiple concurrent users efficiently.

Security:

- Implement role-based authentication using JWT for secure access.
- Encrypt user data to protect personal and professional information.
- Secure API endpoints to prevent unauthorized data access.

Usability:

- Design a clean and intuitive UI using React.js and Tailwind CSS.
- Ensure accessibility for users of all technical backgrounds.

Scalability:

- Use cloud-based deployment to accommodate increasing users and job listings.
- Optimize database queries in MySQL to maintain fast performance as data grows.

Availability:

- Ensure 24/7 access to job listings, mentorship features, and networking opportunities.
- Implement backup and failover strategies for continuous service uptime.

2.8. SUMMARY

The System is internally connected with students, mentors, and job platforms. It can be accessed by authorized students and mentors and allows students to manage their profiles and skills. Students will fill out a profile and add their skills, such as Python, C, etc., in the dashboard. Based on the listed skills, the system will fetch real-time job openings from platforms like LinkedIn and Glassdoor, helping freshers find suitable job opportunities. The existing system has some drawbacks, such as a lack of skill-based job recommendations, limited interaction with mentors, and difficulties in networking with peers. Additionally, the absence of real-time job updates and resource recommendations poses challenges for students. The current system overview includes a website that enables students to create profiles but lacks dynamic job listings and professional guidance.

The proposed system enhances real-time job matching based on students' skills. It introduces immediate job notifications for students, mentor interaction features, and the ability for students to request specific resources. The system also enables students to connect and build networks with other students. Students create profiles with necessary information, add skills, and receive job recommendations. Mentors provide professional guidance and suggest learning resources based on student requests. The software environment describes the desired features from the perspectives of students and mentors.

Functional requirements include user registration, skill management, job matching, mentor interactions, and real-time notifications. Non-functional requirements cover performance, security, usability, scalability, and availability. Overall, the proposed system aims to address the limitations of the existing system by introducing real-time job updates, enhancing user experience, and improving accessibility and efficiency in job searching, mentorship, and professional networking for students.

3. SYSTEM DESIGN

3.1 INPUT DESIGN

Input design for the job search and mentorship application involves creating user interfaces and mechanisms for students to efficiently input their information. This includes designing forms for students to provide their personal details, educational background, and skills. It also involves implementing features that allow students to search for job opportunities, request mentorship, and network with peers. The design should prioritize simplicity, clarity, and accessibility to ensure smooth user interaction and encourage engagement with the platform.

Forms and Interfaces: Designing user-friendly forms and interfaces for students to input their personal information, educational details, and skills. This includes fields for skills such as Python, Java, and C, allowing students to customize their profiles based on their expertise. The system will also provide structured input options for requesting mentorship and accessing career resources.

Validation: Implementing validation checks to ensure the accuracy and completeness of user inputs. This includes validating email addresses, phone numbers, and skill inputs to prevent errors and improve data quality. Additionally, real-time validation will be applied for structured fields like job applications and mentorship requests to ensure relevant information is provided.

Job Search & Recommendations: Allowing students to input their preferred job roles, locations, and industries. The system will use this information, along with listed skills, to fetch real-time job openings from platforms like LinkedIn and Glassdoor, ensuring relevant job recommendations.

Mentorship Requests: Providing an intuitive interface for students to request mentorship by selecting specific areas of interest or challenges they face. Mentors will be able to view these requests and respond with professional guidance and resources.

Networking Features: Enabling students to input their interests and connect with peers who share similar career goals. This will facilitate networking, collaboration, and knowledge sharing among students.

3.2 OUTPUT DESIGN

The output design ensures clear, relevant, and user-friendly information presentation to enhance job searching, mentorship, and networking.

1. **Job Listings & Recommendations:** Display real-time job openings based on users' skills from platforms like LinkedIn and Glassdoor.
2. **Mentorship Notifications:** Notify users about mentor responses, resource availability, and professional guidance updates.
3. **Networking Updates:** Provide suggestions for connecting with peers who share similar career interests.
4. **Skill-Based Insights:** Display personalized career recommendations and growth opportunities based on user-entered skills.

3.3 ARCHITECTURE DIAGRAM

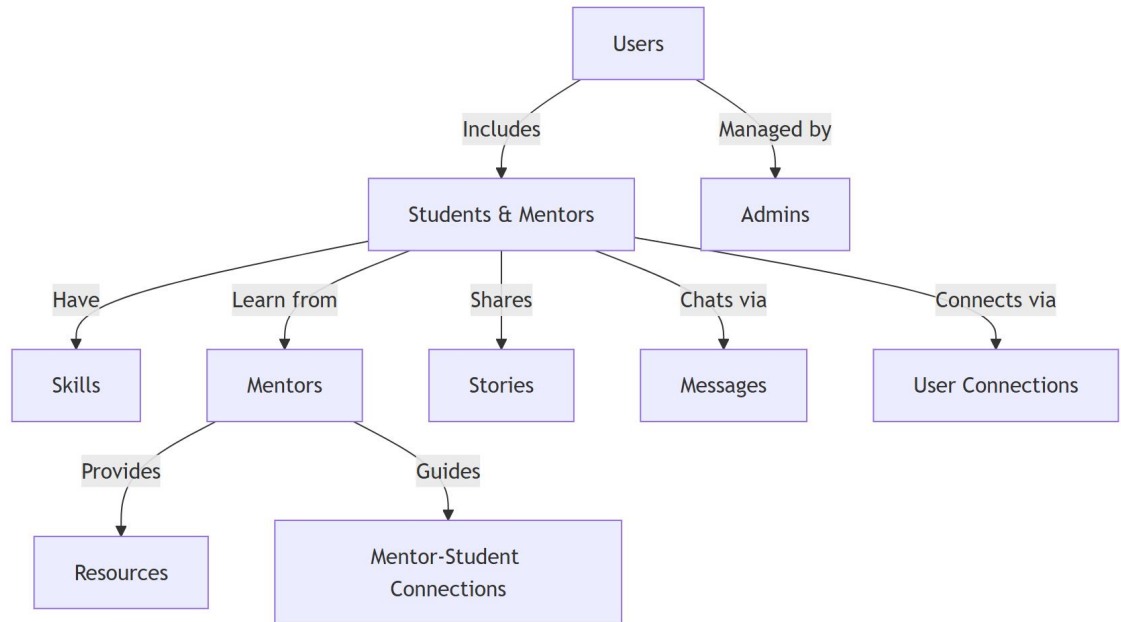


Figure 3.3 ARCHITECTURE DIAGRAM

The blood donation application involves defining various components and their interactions within the system. Here's a high level overview of the typical components you might find in the application

3.3.1. COMPONENTS AND FLOW

The platform is designed to facilitate interactions between students, mentors, and administrators, ensuring a structured and seamless learning experience. At its core, the system revolves around users, who are categorized as students, mentors, and administrators. Students and mentors form the primary interactive group, where students seek guidance and mentors provide knowledge and support. Administrators oversee the entire system, ensuring smooth operations, user management, and maintaining platform integrity.

- **Users** – The core entity, including students, mentors, and admins.
- **Students & Mentors** – A subcategory of users who engage in learning, teaching, and sharing knowledge.

- **Admins** – Responsible for managing platform activities, moderating user interactions, and ensuring smooth functionality.
- **Skills** – Attributes that students and mentors possess, which are essential for learning and mentorship.
- **Mentors** – Experienced individuals who provide guidance, share insights, and help students improve.
- **Resources** – Study materials, guides, and other learning aids shared by mentors to help students.
- **Mentor-Student Connections** – A structured mentorship system where mentors guide students through their learning journey.
- **Stories** – Personal experiences and success stories shared by students and mentors to inspire and educate others.
- **Messages** – A communication system that allows users to chat and interact with each other.
- **User Connections** – A feature enabling networking between users for collaboration and engagement.

3.3.2. PROCESS FLOW

The system connects students and mentors for learning and guidance. Mentors provide resources, and students showcase skills and share experiences. Admins manage operations to ensure efficiency.

1. User Access and Interactions

Users include both **Students** and **Mentors**. They:

- Register and manage profiles that showcase their **Skills**.
- Use the system to **connect** and **communicate** via:
 - I. **Messages**: To chat with each other.
 - II. **User Connections**: To build mentorship relationships.

2. Mentorship and Learning Process

Students:

- Learn and receive guidance from **Mentors**.

- Share **Stories** that capture their learning experiences.

Mentors:

- Provide **Resources** to aid students.
- Facilitate and guide **Mentor-Student Connections** for effective interaction.

3. Admin Role

The entire system is supervised and managed by **Admins**, ensuring smooth operations and user management.

3.4 USE CASE DIAGRAM

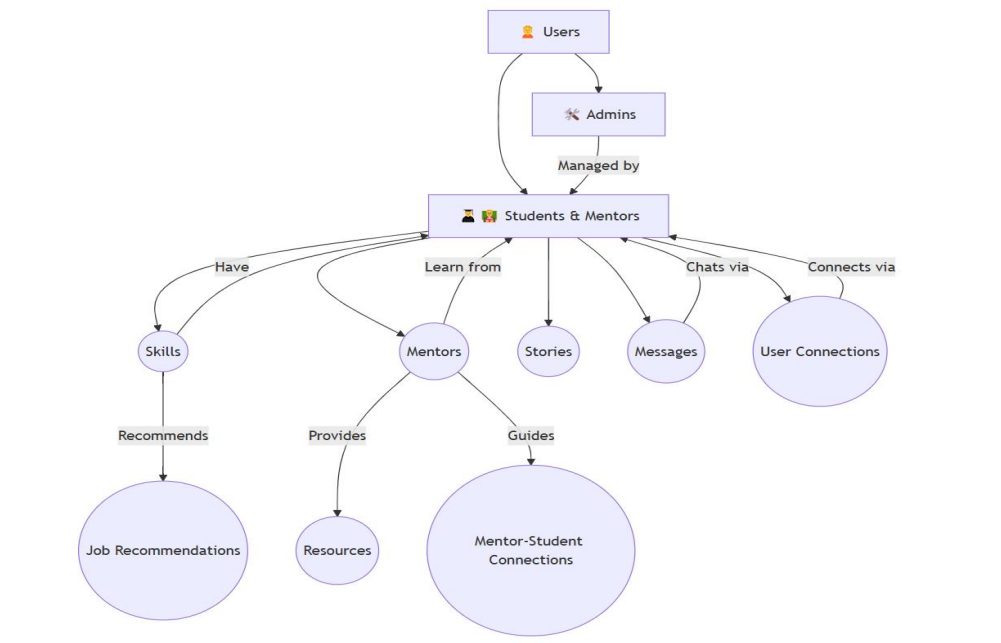
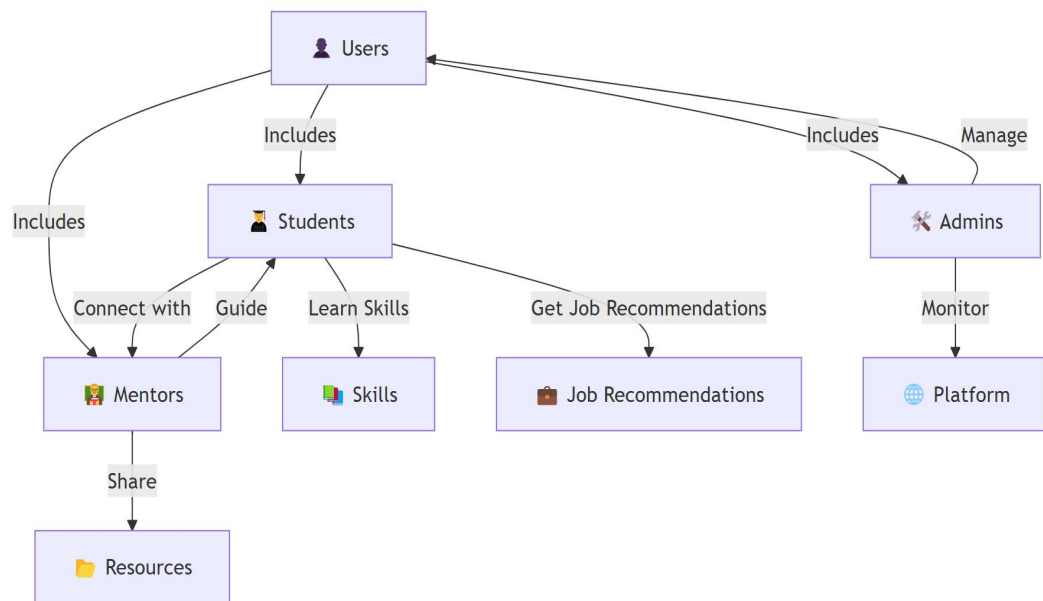


Figure 3.4 USE CASE DIAGRAM

The Use Case Diagram for a job matching system, depicting the interactions between a user and various functionalities of the system

3.4.1. ACTORS

- **Users** – Represents all users of the system, including students, mentors, and admins.
- **Admins** – Responsible for managing students and mentors.
- **Students & Mentors** – The core participants who engage in various activities such as learning, mentoring, and skill development.



3.4.2. USE CASES

- ✓ **Skills** – Students possess skills that define their expertise and learning areas.
- ✓ **Mentors** – Experienced users who guide students and provide learning resources.
- ✓ **Stories** – A platform where experiences, success stories, and knowledge are shared.
- ✓ **Messages** – Allows users to communicate and interact within the system.
- ✓ **User Connections** – Establishes relationships and networking between users.
- ✓ **Resources** – Learning materials provided by mentors to help students develop skills.
- ✓ **Mentor-Student Connections** – A structured relationship where mentors guide students in their learning journey.
- ✓ **Job Recommendations** – The system recommends relevant jobs to students based on their acquired skills.

3.4.3. RELATIONSHIPS

1. Users → Students & Mentors (Generalization)

All users include students and mentors as key participants.

2. Admins → Students & Mentors (Manages)

Admins oversee and manage student and mentor activities.

3. Students & Mentors → Skills (Have)

Students acquire and develop skills, forming the foundation for job recommendations.

4. Skills → Job Recommendations (Recommends)

Once students add skills, the system recommends job opportunities relevant to their expertise.

5. Students & Mentors → Mentors (Learn From)

Students seek guidance and knowledge from mentors.

6. Mentors → Resources (Provides)

Mentors create and share learning resources.

7. Mentors → Mentor-Student Connections (Guides)

Mentors actively guide students through structured mentorship.

8. Students & Mentors → Stories (Shares)

Users share their experiences, projects, and success stories.

9. Students & Mentors → Messages (Chats Via)

Users communicate with each other through messaging.

10. Students & Mentors → User Connections (Connects Via)

Users establish professional and learning connections

3.4.4 Flow Summary:

1. A student registers and starts adding skills.
2. Based on acquired skills, the system recommends suitable jobs.
3. Students connect with mentors for guidance and resources.
4. Mentors provide learning materials and engage in mentor-student connections.
5. Users share their experiences through stories.
6. Messaging enables communication, and user connections allow networking.
7. Admins oversee the overall system and user interactions.

3.5 Class Diagram

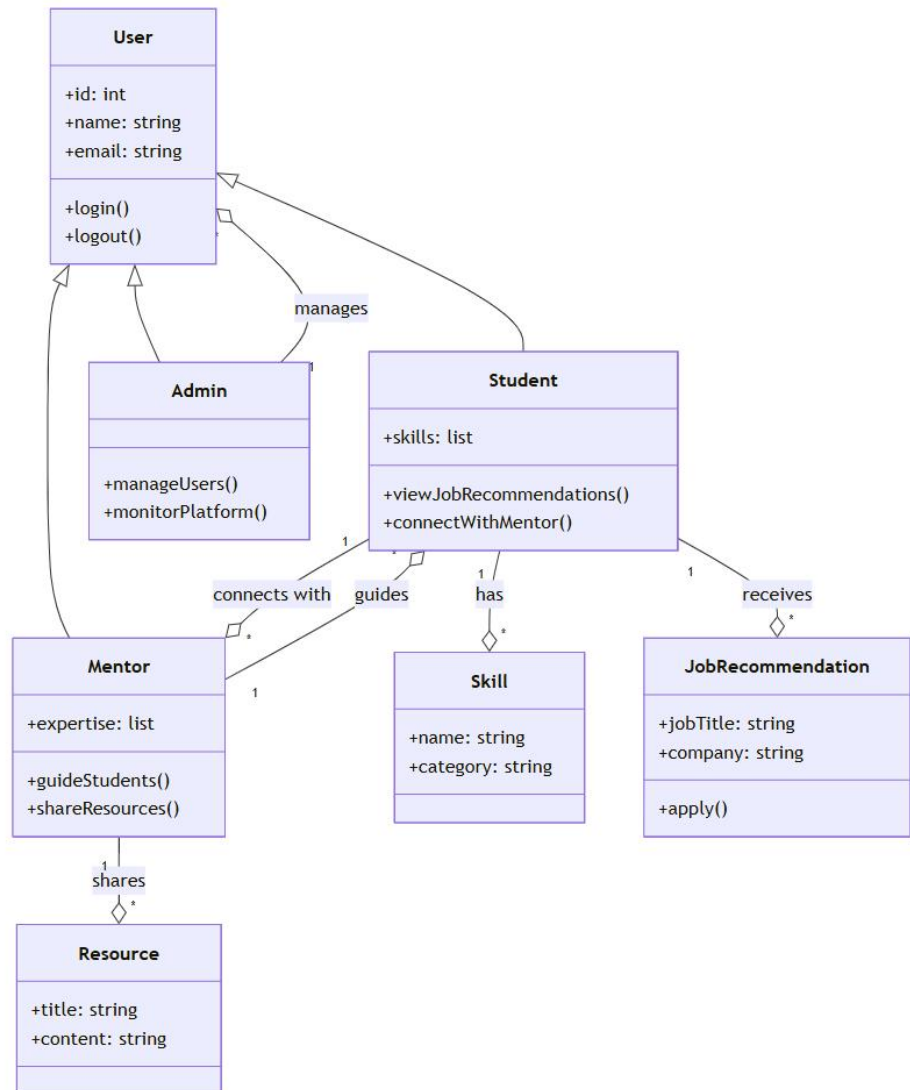


Figure 3.5 CLASS DIAGRAM

1. USER

1.1 Attributes

- `+id`: Unique identifier for the user.
- `+name`: Name of the user.
- `+email`: User's email address.

-

1.2 Methods

- +login(): Authenticates the user.
- +logout(): Logs the user out.

1.3 Relationships

- **User is a general class** that is inherited by:
 - Student (Has skills, connects with mentors, and receives job recommendations).
 - Mentor (Provides guidance and shares resources).
 - Admin (Manages users).

2. STUDENT

2.1 Attributes

- +skills: List of skills acquired by the student.

2.2 Methods

- +viewJobRecommendations(): Allows students to view job recommendations based on their skills.
- +connectWithMentor(): Enables students to connect with mentors for guidance.

2.3 Relationships

- Students have multiple skills.
- Students receive job recommendations based on their skills.
- Students connect with mentors for guidance.
-

3. MENTOR

3.1 Attributes

- +expertise: List of expertise areas of the mentor.

3.2 Methods

- +guideStudents(): Provides guidance and mentorship to students.
- +shareResources(): Shares educational resources with students.

3.3 Relationships

- Mentors guide students.
- Mentors share resources.

4. ADMIN

4.1 Methods

- +manageUsers(): Admins can manage users, including students and mentors.
- +monitorPlatform(): Admins can monitor activities within the platform.

4.2 Relationships

- Admins manage users (students and mentors).

5. SKILL

5.1 Attributes

- +name: The name of the skill.
- +category: The category to which the skill belongs.

5.2 Relationships

- Students have multiple skills.

6. JOB RECOMMENDATION

6.1 Attributes

- +jobTitle: The title of the recommended job.
- +company: The company offering the job.

6.2 Methods

- +apply(): Allows students to apply for job recommendations.

6.3 Relationships

- Students receive job recommendations based on their skills.

7. RESOURCE

7.1 Attributes

- +title: Title of the resource.
- +content: The content of the resource.

7.2 Relationships

- Mentors share resources with students.

3.6 ACTIVITY DIAGRAM

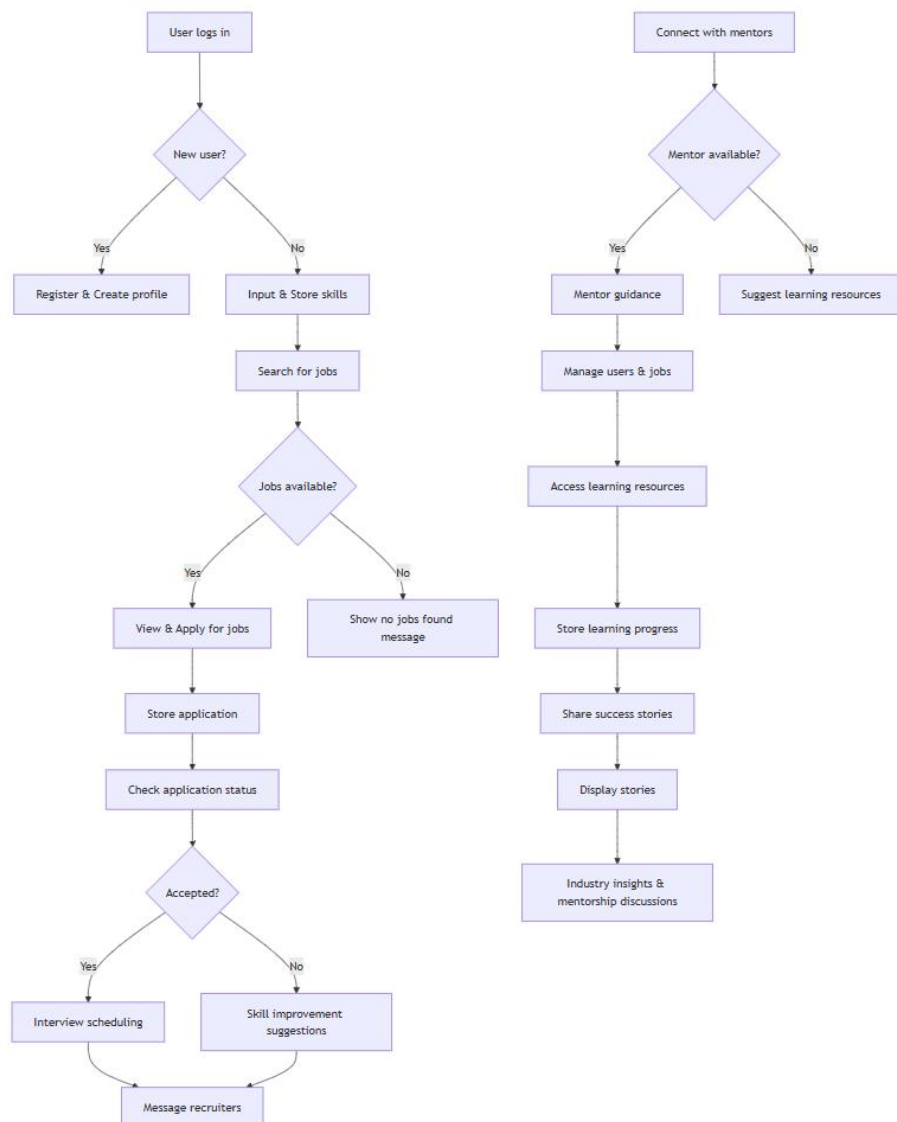


Figure 3.6 ACTIVITY DIAGRAM

1. Login Process

- The process starts with a **login action**.
- If the login is **successful**, the user proceeds to **ChooseRole**.
- If the login **fails**, the process terminates.

2. Role Selection (ChooseRole)

- The user selects a role from **Student, Mentor, or Admin**.

3. Student Role

- The student can **AddSkills**.
- After adding skills, they have two options:
 - **ViewJobRecommendations** → Leads to **ApplyForJob**.
 - **ConnectWithMentor** → Leads to **ReceiveGuidance**.
- Once a student either applies for a job or receives guidance, the process terminates.

4. Mentor Role

- The mentor can **GuideStudents**.
- They can also **ShareResources**.
- After sharing resources or guiding students, the process ends.

5. Admin Role

- The admin can **ManageUsers**.
- They can also **MonitorPlatform**.
- Once they complete these tasks, the process ends

3.7. ER DIAGRAM

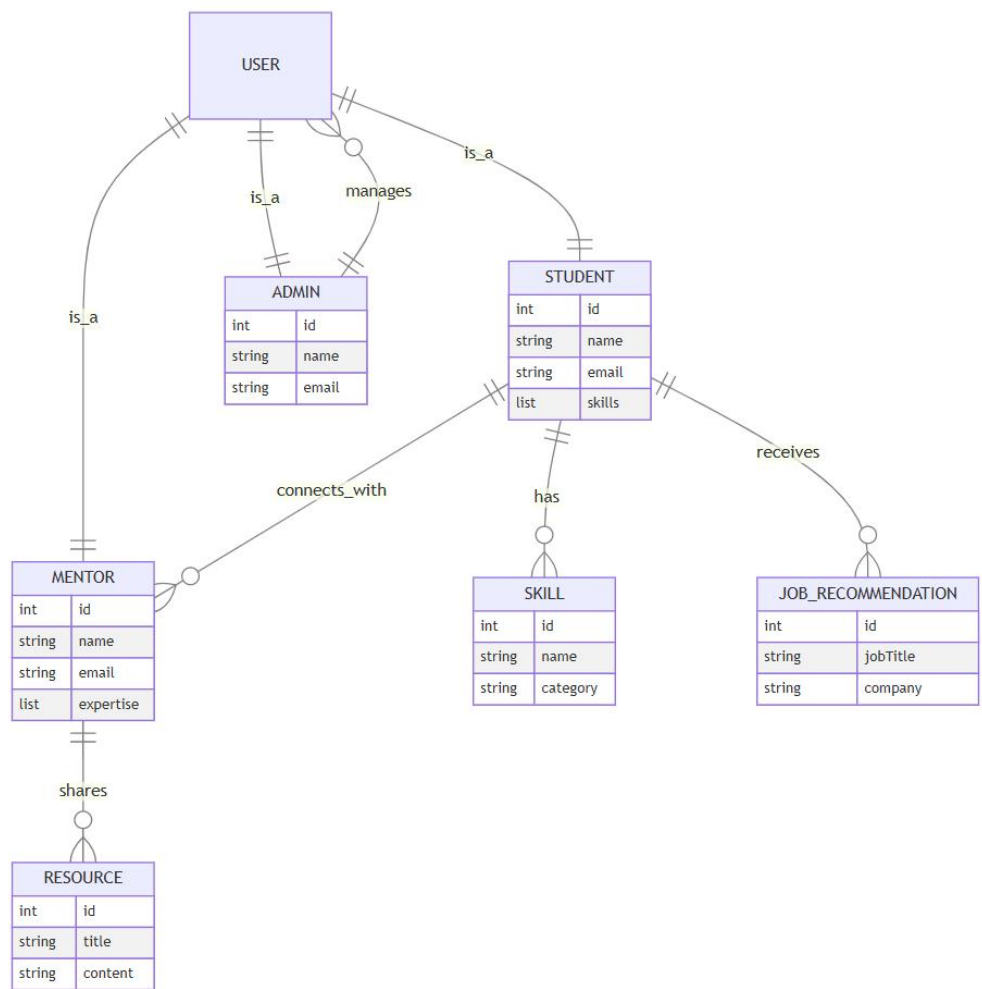


Figure 3.7 ER DIAGRAM

The ER (Entity-Relationship) diagram represents a **mentorship and job recommendation platform**. Below is a detailed breakdown of the entities, attributes, and relationships in the system.

Entities and Attributes:

1. User (Generalized Entity)

1. Attributes:
 1. id (Unique Identifier)
 2. name (User's Name)

3. email (User's Email)
- 4.
2. Relationships:
 1. Generalized into three roles: **Student, Mentor, and Admin.**
2. **Student (Inherits from User)**
 1. Attributes:
 1. skills (List of skills the student possesses)
 2. Relationships:
 1. **Has Skills:** A Student possesses multiple skills.
 2. **Receives Job Recommendations:** A Student gets job recommendations.
 3. **Connects with Mentor:** A Student can connect with a mentor for guidance.
3. **Mentor (Inherits from User)**
 1. Attributes:
 1. expertise (List of areas where the mentor has expertise)
 2. Relationships:
 1. **Guides Students:** A Mentor helps students by offering guidance.
 2. **Shares Resources:** A Mentor can share educational or professional resources.
4. **Admin (Inherits from User)**
 1. Attributes:
 1. (No additional attributes, inherits basic User attributes)

2. Relationships:

1. **Manages Users:** An Admin is responsible for managing students and mentors.

5. Skill

1. Attributes:

1. name (Name of the skill)
2. category (Category of the skill)

2. Relationships:

1. **Belongs to Student:** A Student can have multiple skills.

6. Job Recommendation

1. Attributes:

1. jobTitle (Title of the recommended job)
2. company (Company offering the job)

2. Relationships:

1. **Sent to Students:** A Student receives job recommendations based on skills.

7. Resource

1. Attributes:

1. title (Name of the resource)
2. content (Description or details of the resource)

2. Relationships:

1. **Shared by Mentor:** A Mentor can share multiple resources.

Relationships and Their Meaning:

- **User is specialized into Student, Mentor, and Admin** → A person using the system can have one of these roles.
- **Student has Skills** → Each student has a set of skills that define their expertise.
- **Student receives Job Recommendations** → Based on their skills, students get job recommendations.
- **Student connects with Mentor** → Students can reach out to mentors for professional guidance.
- **Mentor guides Students** → Mentors provide guidance and career advice to students.
- **Mentor shares Resources** → Mentors share useful materials with students to enhance their learning.
- **Admin manages Users** → Admins oversee user management and maintain system integrity.

General Flow of the System:

1. **Users register** and select their role as **Student, Mentor, or Admin**.
2. **Students add skills** to their profiles.
3. **Students receive job recommendations** based on their skills.
4. **Students can connect with Mentors** for career guidance.
5. **Mentors provide guidance and share resources** to help students improve their skills.
6. **Admins oversee the platform**, managing users and ensuring smooth operation.

3.8 SUMMARY

1. Architecture Overview

The architecture of the mentorship and job recommendation platform is designed to facilitate seamless interactions between students, mentors, and administrators. It consists of the following layers and components:

User Interface (UI):

- Provides an intuitive platform for students, mentors, and admins to interact.
- Enables functionalities such as profile creation, mentorship connections, job recommendations, and resource sharing.

Backend Services:

- **Database:** Stores user information, skills, job recommendations, mentorship connections, and resources.
- **User Authentication:** Manages login, registration, and access control.

Role-Based Access Control:

- **Students:** Can view job recommendations, connect with mentors, and apply for jobs.
- **Mentors:** Can guide students and share educational resources.
- **Admins:** Manage user accounts, monitor platform activity, and oversee mentorship interactions.

2. Components and Workflows

User Registration and Authentication:

- **Registration:** Users register as students, mentors, or admins.
- **Login:** Users authenticate to access platform features.
- **Profile Management:** Users update their details, such as skills, expertise, and preferences.

Mentorship Process:

- **Mentor Connection:** Students can browse and connect with mentors based on their expertise.
- **Guidance and Support:** Mentors provide career guidance and share industry insights.
- **Resource Sharing:** Mentors upload and distribute learning materials to students.

Job Recommendation Process:

- **Skill Matching:** The system suggests job opportunities based on student skills.
- **Application Process:** Students can apply for jobs directly through the platform.

Admin Management:

- **User Moderation:** Admins monitor users and ensure a safe environment.
- **System Oversight:** Admins handle platform-wide settings, analytics, and issue resolution.

3. Use Cases and User Interactions

The system enables various user interactions:

- **Register/Login:** Users create accounts and sign in to access the platform.
- **Edit Profile:** Students and mentors update their information, including skills and expertise.
- **Find Mentors:** Students search for and connect with relevant mentors.
- **View Job Recommendations:** Students receive job suggestions based on their skills.
- **Apply for Jobs:** Students can directly apply for recommended job opportunities.
- **Guide Students:** Mentors provide career advice and support.
- **Share Resources:** Mentors upload educational materials for students.
- **Manage Users:** Admins oversee platform activity and user engagement.

4. Class Diagram (Inferred)

The class diagram defines the key components of the system:

- **User:** Base class for all users with methods for registration and authentication.
 - **Student:** Inherits from User, with additional attributes like skills and methods for viewing job recommendations and connecting with mentors.
 - **Mentor:** Inherits from User, with attributes like expertise and methods for guiding students and sharing resources.
 - **Admin:** Oversees system management and user monitoring.
- **Skill:** Represents various skills associated with students.
- **JobRecommendation:** Contains job details such as jobTitle and company, associated with students.
- **Resource:** Represents educational materials shared by mentors.

Conclusion

The mentorship and job recommendation platform streamlines career growth by connecting students with mentors, providing job opportunities, and facilitating skill development. With an intuitive interface, real-time notifications, and role-based access, the system ensures seamless interactions for students, mentors, and admins. This comprehensive solution enhances professional guidance and employment prospects, fostering a collaborative learning environment.

4. SYSTEM IMPLEMENTATION

4.1. INTRODUCTION:

The development and implementation of a mentorship and job recommendation platform represent a significant advancement in career guidance and professional networking. This platform addresses the critical need for a structured, user-friendly system that connects students, mentors, and job opportunities seamlessly. This section provides an overview of the system implementation, detailing its technical architecture, core components, and key functionalities.

The platform is built on a robust, scalable framework that ensures reliability and efficiency, leveraging cloud-based storage for real-time data access and seamless communication. The application integrates advanced recommendation algorithms, allowing students to receive job suggestions based on their skills and interests while enabling mentors to provide targeted guidance. These features enhance career progression and knowledge-sharing among users.

The mentorship system facilitates structured interactions between students and mentors through personalized recommendations, scheduled meetings, and knowledge-sharing tools. Mentors can guide students through skill development, career planning, and industry insights, ensuring meaningful professional growth.

A key feature of the platform is its intelligent notification system, which alerts students to new job recommendations, mentor availability, and relevant skill-building opportunities. These notifications, tailored to individual user profiles, enhance engagement and encourage proactive career development. Additionally, the platform provides educational resources, including career-related articles, industry trends, and skill-enhancement materials, fostering continuous learning.

Initial deployment of the platform has demonstrated increased student engagement, mentor participation, and successful job placements.

4.2. MODULES:

4.2.1. Module 1: User Interface (UI)

The User Interface (UI) module of the web application ensures a seamless and intuitive experience for users, including students, mentors, and administrators. It is built using modern frontend technologies such as **HTML, CSS, JavaScript, and React.js** to provide a responsive and interactive interface.

Key components of the UI module:

1. **Layouts:** Defines the structure and visual appearance of pages, including dashboards, job recommendations, mentorship connections, and admin panels.
2. **Navigation & Routing:** Implements smooth navigation between pages using React Router or other routing frameworks.
3. **Forms & Dialogs:** Interactive forms for user registration, job applications, mentor requests, and admin approvals.
4. **Responsive Design:** Ensures adaptability across different devices using frameworks like Bootstrap or Tailwind CSS.

This module enhances user experience by providing a clean, structured, and efficient interface for interacting with the system.

4.2.2. Module 2: User Management

The User Management module handles authentication, authorization, and user profiles. It ensures secure access and role-based functionalities for different types of users.

Key features include:

1. **User Registration & Login:**
 1. Users (students, mentors, admins) register using email, social logins, or other authentication methods.
 2. Secure login using JWT (JSON Web Token) authentication.

2. Role-Based Access Control:

1. Students can view job recommendations and connect with mentors.
2. Mentors can guide students and share resources.
3. Admins manage users, monitor platform activity, and approve mentorship requests.

3. Profile Management:

1. Users can edit their profiles, update skills, and manage settings.
2. Secure password management and account recovery options.

4.2.3. Module 3: Job and Mentoring System

This module is responsible for connecting students with mentors and job opportunities based on their skills and interests.

Key features:

1. Job Recommendation System:

1. Uses AI-based algorithms or predefined filters to suggest job opportunities based on a student's skills and preferences.
2. Employers can list job openings and receive applications from suitable candidates.

2. Mentorship Matching:

1. Students can find mentors based on expertise, interests, and industry experience.
2. Mentors can approve or decline requests and schedule one-on-one sessions.

3. Search & Filtering:

1. Users can search for job openings or mentors using filters like industry, skill set, and location.

4.2.4. Module 4: Admin Dashboard

The Admin Dashboard allows administrators to oversee platform operations, manage users, and maintain system integrity.

Key features:

1. User Management:

1. View and manage registered users, including students, mentors, and companies.
2. Approve or reject mentor applications.

2. Job & Mentor Oversight:

1. Monitor job postings and mentorship activities.
2. Ensure compliance with platform guidelines.

3. Analytics & Reporting:

1. View platform statistics, including user engagement, mentorship success rates, and job placements.
2. Generate reports for insights into system performance.

4.2.5. Module 5: Community Engagement

The Community Engagement module fosters interaction, collaboration, and knowledge-sharing among students, mentors, and employers. It helps build a strong, supportive ecosystem where users can engage in discussions, share experiences, and contribute to the platform's growth.

Key Features:

1. Discussion Forums & Q&A:

- Users can participate in topic-based forums to discuss industry trends, skills, career guidance, and job opportunities.
- Mentors can provide expert advice, and students can ask questions about

career paths, internships, and projects.

2. Project Collaboration:

- Students and mentors can form groups to work on projects, exchange ideas, and gain practical experience.
- Employers can observe talented individuals and offer internship or job opportunities.

3. Resource Sharing:

- Users can share study materials, research papers, articles, and tutorials related to various domains.
- A repository of learning resources helps students upskill and stay updated with industry demands.

4.3 IMPLEMENTATION PROCEDURE

1. Requirements Analysis

Gather requirements:

- Collect and document functional and non-functional requirements from key stakeholders, including students, job seekers, mentors, and employers.
- Define core functionalities such as job searching, career counseling, resume building, and skill assessment.

Feasibility study:

- Assess the technical, operational, and financial feasibility of the application.
- Evaluate cloud hosting options, database management systems, and front-end frameworks for scalability and security.

2. System Design

Architecture design:

- Define the overall architecture, selecting technologies like React.js for the front end, Node.js for the back end, and MySQL for data storage.

Database design:

- Design the database schema to store user profiles, job postings, mentor details, application statuses, and career resources.
- Ensure data normalization, security measures, and efficient indexing.

User interface and user experience design:

- Create wireframes and mockups to define a seamless user experience.
- Design intuitive dashboards for students, mentors, and employers, ensuring accessibility and ease of navigation.

3. Development

Set up development environment:

- Configure development environments, including version control (Git or GitHub), cloud services, and continuous integration and continuous deployment pipelines.

Front-end development:

- Implement the user interface using React.js and Tailwind CSS for a responsive, modern look.
- Develop reusable components for job listings, mentorship sessions, and profile management.

Back-end development:

- Develop the server-side logic using Node.js with Express.js.

- Implement representational state transfer application programming interfaces to manage user authentication, job listings, applications, and mentorship sessions.

Integration of modules:

To develop and integrate the various modules:

- ✓ User interface module
- ✓ User management module
- ✓ Job search and application module
- ✓ Mentorship and career guidance module
- ✓ Community engagement module

4. Testing**Unit testing:**

- Conduct unit tests for individual components, such as React components, application programming interface endpoints, and database operations.

Integration testing:

- Test the interaction between the front end and back end, ensuring seamless data flow.

System testing:

- Perform end-to-end testing of all features, verifying responsiveness and real-time data updates.

User acceptance testing:

- Conduct user acceptance testing with students, mentors, and employers to gather feedback and refine functionalities.

5. Deployment

Prepare deployment environment:

- Set up cloud-based hosting, such as Amazon Web Services, Firebase, or Vercel, and configure production servers.
- Optimize database queries and application programming interface performance for scalability.

Deploy application:

- Deploy the application using Vercel for the front end and Amazon Web Services or Firebase for the back end and database.
- Ensure security configurations, such as hypertext transfer protocol secure encryption and authentication protocols.

Post-deployment testing:

- Conduct live testing in the production environment to ensure smooth performance and resolve any last-minute issues.

4.4 SOURCE CODE

Source Code Structure and Organization

The **Narayana Career Connect** web application is structured into several directories and files, each serving a specific role in the development, functionality, and operation of the app. Below is an overview of the main directories and files in the project:

Main Directories and Files**1. src/ (Source Code - Frontend & Backend)**

This directory contains all the frontend and backend source code files, defining the application's logic and user interface.

client/ (Client-side code)

- components/ → React/Angular/Vue components for UI elements.

- pages/ → Individual pages of the application (e.g., Home, Job Listings, Profile).
- styles/ → CSS/SCSS/Tailwind files for styling.
- utils/ → Helper functions and utilities.
- App.js / index.js → Main entry point for rendering the frontend.

server/ (Server-side code)

- routes/ → API routes for user authentication, job listings, applications, etc.
- controllers/ → Logic for handling API requests.
- models/ → Database models (e.g., User, Job, Application).
- middleware/ → Authentication and validation logic.
- server.js → Main entry file for the backend (Node.js/Express).

CODE:

1. src/pages/Register.jsx

```
import React, { useState } from 'react';  
import axios from 'axios';  
import { useNavigate } from 'react-router-dom';
```

```
function Register() {  
  const [formData, setFormData] = useState({  
    name: "",  
    password: "",  
    email: "",  
    role: 'student',
```

```
    bio: "",
    is_past_student: false,
    current_year: "",
    current_sem: "",
    profile_pic: null,
    college: 'NECN',
  });

const [errors, setErrors] = useState({});
const [serverMessage, setServerMessage] = useState(null);
const [isSubmitting, setIsSubmitting] = useState(false);
const navigate = useNavigate();

const handleChange = (e) => {
  const { name, value, type, checked, files } = e.target;
  if (type === 'file') {
    setFormData({ ...formData, [name]: files[0] });
  } else if (type === 'checkbox') {
    setFormData({ ...formData, [name]: checked });
  } else {
    setFormData({ ...formData, [name]: value });
  }
};

const handleSubmit = async (e) => {
  e.preventDefault();
  setIsSubmitting(true);

  const formPayload = new FormData();
  Object.entries(formData).forEach(([key, value]) => {
    formPayload.append(key, value);
  });
};
```

```

    try {
      const response = await axios.post('http://localhost:3000/api/auth/register',
formPayload, {
        headers: { 'Content-Type': 'multipart/form-data' },
      });

      if (response.data && response.data.message) {
        setServerMessage({ type: 'success', text: response.data.message });
      } else {
        setServerMessage({ type: 'error', text: 'Registration failed. Please try again.' });
      }

      if (response.data.success) {
        setTimeout(() => navigate('/login'), 1500);
      }
    } catch (error) {
      if (error.response && error.response.data.message) {
        setErrors({ server: error.response.data.message });
      } else {
        setErrors({ server: 'An unexpected error occurred. Please try again.' });
      }
    } finally {
      setIsSubmitting(false);
    }
  };

  return (
    <div className="min-h-screen bg-gradient-to-br from-indigo-50 to-blue-100 flex
items-center justify-center p-4">
      <div className="w-full max-w-2xl bg-white rounded-xl shadow-lg overflow-
hidden">

```

```

<div className="flex flex-col md:flex-row">
  {/ * Left sidebar/accent area - visible on md screens and up */}
  <div className="hidden md:block md:w-1/3 bg-gradient-to-b from-blue-
500 to-indigo-600 p-8">
    <div className="h-full flex flex-col justify-between">
      <div>
        <h2 className="text-2xl font-bold text-white mb-4">Join Our
Community</h2>
        <p className="text-blue-100 mb-8">Create an account to connect
with mentors and access learning resources.</p>
      </div>
      <div className="text-blue-200 text-sm">
        <p>Already have an account?</p>
        <a href="/login" className="text-white font-bold hover:underline
mt-1 inline-block">
          Sign in here
        </a>
      </div>
    </div>
  </div>

  {/ * Main content area */}
  <div className="w-full md:w-2/3 p-6 md:p-8">
    <div className="mb-8">
      <h1 className="text-2xl font-bold text-gray-800">Create an
Account</h1>
      <p className="text-gray-600 mt-1">Fill in your details to get
started</p>

      {/ * Mobile login link */}
      <div className="block md:hidden mt-4 text-sm">
        <span className="text-gray-500">Already have an

```


account?

<a href="/login" className="text-blue-600 font-medium
hover:underline ml-1">

Sign in

</div>

</div>

{serverMessage && (

<div className={`p-4 rounded-lg mb-6 \${

serverMessage.type === 'success'

? 'bg-green-50 text-green-700 border border-green-200'

: 'bg-red-50 text-red-700 border border-red-200'

}`}>

<p className="flex items-center">

{serverMessage.type === 'success' ? (

<svg className="w-5 h-5 mr-2" fill="currentColor"

viewBox="0 0 20 20" xmlns="http://www.w3.org/2000/svg">

<path fillRule="evenodd" d="M10 18a8 8 0 10-16 8 8 0
000 16zm3.707-9.293a1 1 0 00-1.414-1.414L9 10.586 7.707 9.293a1 1 0 00-1.414
1.414l2 2a1 1 0 001.414 0l4-4z" clipRule="evenodd"></path>

</svg>

): (

<svg className="w-5 h-5 mr-2" fill="currentColor"

viewBox="0 0 20 20" xmlns="http://www.w3.org/2000/svg">

<path fillRule="evenodd" d="M18 10a8 8 0 11-16 0 8 8 0
0116 0zm-7 4a1 1 0 11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-
1-1z" clipRule="evenodd"></path>

</svg>

))

{serverMessage.text}

</p>

```

    </div>
  })

  {errors.server && (
    <div className="p-4 rounded-lg mb-6 bg-red-50 text-red-700 border
border-red-200">
      <p className="flex items-center">
        <svg className="w-5 h-5 mr-2" fill="currentColor"
viewBox="0 0 20 20" xmlns="http://www.w3.org/2000/svg">
          <path fillRule="evenodd" d="M18 10a8 8 0 11-16 0 8 8 0
0116 0zm-7 4a1 1 0 11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-
1-1z" clipRule="evenodd"></path>
        </svg>
        {errors.server}
      </p>
    </div>
  })

  <form onSubmit={handleSubmit} className="space-y-5">
    <div className="grid grid-cols-1 md:grid-cols-2 gap-5">
      <div>
        <label className="block text-sm font-medium text-gray-700
mb-1">Full Name</label>
        <input
          type="text"
          name="name"
          value={formData.name}
          onChange={handleChange}
          placeholder="Enter your full name"
          className={`w-full px-4 py-2 rounded-lg border
${errors.name ? 'border-red-500' : 'border-gray-300'} focus:outline-none focus:ring-2
focus:ring-blue-500 focus:border-transparent transition`}

```

```

        />
        {errors.name && <p className="mt-1 text-sm text-red-
600">{errors.name}</p>}
    </div>

    <div>
        <label className="block text-sm font-medium text-gray-700
mb-1">Email Address</label>
        <input
            type="email"
            name="email"
            value={formData.email}
            onChange={handleChange}
            placeholder="your.email@example.com"
            className={`w-full px-4 py-2 rounded-lg border
${errors.email ? 'border-red-500' : 'border-gray-300'} focus:outline-none focus:ring-2
focus:ring-blue-500 focus:border-transparent transition`}
        />
        {errors.email && <p className="mt-1 text-sm text-red-
600">{errors.email}</p>}
    </div>
</div>

    <div>
        <label className="block text-sm font-medium text-gray-700 mb-
1">Password</label>
        <input
            type="password"
            name="password"
            value={formData.password}
            onChange={handleChange}
            placeholder="Create a secure password"

```

```

        className={`w-full px-4 py-2 rounded-lg border
        ${errors.password ? 'border-red-500' : 'border-gray-300'} focus:outline-none focus:ring-
        2 focus:ring-blue-500 focus:border-transparent transition`}
      />
      {errors.password} <p className="mt-1 text-sm text-red-
600">{errors.password}</p>
    </div>

    <div className="grid grid-cols-1 md:grid-cols-2 gap-5">
      <div>
        <label className="block text-sm font-medium text-gray-700
mb-1">Role</label>
        <select
          name="role"
          value={formData.role}
          onChange={handleChange}
          className="w-full px-4 py-2 rounded-lg border border-gray-
300 bg-white focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-
transparent transition"
        >
          <option value="student">Student</option>
          <option value="mentor">Mentor</option>
        </select>
      </div>

      <div>
        <label className="block text-sm font-medium text-gray-700
mb-1">College</label>
        <select
          name="college"
          value={formData.college}
          onChange={handleChange}

```

```

        className="w-full px-4 py-2 rounded-lg border border-gray-
300 bg-white focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-
transparent transition"
    >
        <option value="NECN">NECN</option>
        <option value="NECG">NECG</option>
    </select>
</div>
</div>

<div>
    <label className="block text-sm font-medium text-gray-700 mb-
1">Bio</label>
    <textarea
        name="bio"
        value={formData.bio}
        onChange={handleChange}
        placeholder="Tell us about yourself"
        rows="3"
        className="w-full px-4 py-2 rounded-lg border border-gray-300
focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-transparent transition"
    />
</div>

{formData.role === 'mentor' && (
    <div className="flex items-center">
        <input
            type="checkbox"
            name="is_past_student"
            id="is_past_student"
            checked={formData.is_past_student}
            onChange={handleChange}

```

```

        className="h-4 w-4 text-blue-600 focus:ring-blue-500
border-gray-300 rounded"
    />
    <label htmlFor="is_past_student" className="ml-2 block text-
sm text-gray-700">
        I am a past student of this institution
    </label>
</div>
)}

{formData.role === 'student' && (
    <div className="grid grid-cols-1 md:grid-cols-2 gap-5">
        <div>
            <label className="block text-sm font-medium text-gray-700
mb-1">Current Year</label>
            <input
                type="number"
                name="current_year"
                value={formData.current_year}
                onChange={handleChange}
                placeholder="e.g., 1, 2, 3, 4"
                className="w-full px-4 py-2 rounded-lg border border-
gray-300 focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-transparent
transition"
            />
        </div>
        <div>
            <label className="block text-sm font-medium text-gray-700
mb-1">Current Semester</label>
            <input
                type="number"
                name="current_sem"

```

```

        value={formData.current_sem}
        onChange={handleChange}
        placeholder="e.g., 1, 2, 3, 4, 5, 6, 7, 8"
        className="w-full px-4 py-2 rounded-lg border border-
gray-300 focus:outline-none focus:ring-2 focus:ring-blue-500 focus:border-transparent
transition"

      />
    </div>
  </div>
)}

<div>
  <label className="block text-sm font-medium text-gray-700 mb-
1">Profile Picture</label>
  <div className="mt-1 flex items-center">
    <span className="inline-block h-12 w-12 rounded-full
overflow-hidden bg-gray-100">
      {formData.profile_pic ? (
        <img
          src={URL.createObjectURL(formData.profile_pic)}
          alt="Profile preview"
          className="h-full w-full object-cover"
        />
      ) : (
        <svg className="h-full w-full text-gray-300"
fill="currentColor" viewBox="0 0 24 24">
          <path d="M24 20.993V24H0v-2.996A14.977 14.977 0
0112.004 15c4.904 0 9.26 2.354 11.996 5.993zM16.002 8.999a4 4 0 11-8 0 4 4 0 018
0z" />
        </svg>
      )}
    </span>
  </div>

```

```

<label
  htmlFor="profile_pic_upload"
  className="ml-5 cursor-pointer bg-white py-2 px-3 border
border-gray-300 rounded-md shadow-sm text-sm leading-4 font-medium text-gray-700
hover:bg-gray-50 focus:outline-none focus:ring-2 focus:ring-offset-2 focus:ring-blue-
500"
>
  Upload
  <input
    id="profile_pic_upload"
    type="file"
    name="profile_pic"
    onChange={handleChange}
    className="sr-only"
  />
</label>
{formData.profile_pic && (
  <span className="ml-2 text-sm text-gray-500">
    {formData.profile_pic.name}
  </span>
)}
</div>
</div>

<button
  type="submit"
  disabled={isSubmitting}
  className="w-full bg-gradient-to-r from-blue-600 to-indigo-600
hover:from-blue-700 hover:to-indigo-700 text-white font-medium py-3 px-4 rounded-lg
shadow-md hover:shadow-lg transition duration-300 ease-in-out focus:outline-none
focus:ring-2 focus:ring-blue-500 focus:ring-opacity-50 flex justify-center items-center"
>

```



```

        {isSubmitting ? (
            <
                <svg className="animate-spin -ml-1 mr-3 h-5 w-5 text-white"
xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
                    <circle className="opacity-25" cx="12" cy="12" r="10"
stroke="currentColor" strokeWidth="4"></circle>
                    <path className="opacity-75" fill="currentColor" d="M4
12a8 8 0 018-8V0C5.373 0 0 5.373 0 12h4zm2 5.291A7.962 7.962 0 014 12H0c0 3.042
1.135 5.824 3 7.938l3-2.647z"></path>
                </svg>
                Processing...
            </>
        ) : (
            'Create Account'
        )}
    </button>
</form>
</div>
</div>
</div>
</div>
);
}

export default Register;

```

Fig: 4.4.1. Register Page

2. src/pages/Jobs.jsx

```
import React, { useState, useEffect } from 'react';
import { MapPin, Building2, Briefcase, ExternalLink, Info, Loader2 } from 'lucide-react';
import Cookies from 'js-cookie';
import useAuth from '../utils/AuthContext';
import { useNavigate } from 'react-router-dom';

const formatText = (text) => {
  if (!text) return "";
  text = text.replace(/\*(.*?)\*/g, '<strong>$1</strong>');
  text = text.replace(/^(.*?)\*/g, '<strong>$1</strong>');
  return text;
};

const Jobs = () => {
```

```
const [jobs, setJobs] = useState([]);
const [loading, setLoading] = useState(true);
const [error, setError] = useState(null);
const [selectedJob, setSelectedJob] = useState(null);
const [lastFetchTime, setLastFetchTime] = useState(null);

const {token} = useAuth();
const navigate = useNavigate();

useEffect(() => {
  if(!token){
    navigate('/')
  }
},[token, navigate]);

useEffect(() => {
  fetchJobs();

  // Set up periodic refresh every 5 minutes
  const refreshInterval = setInterval(() => {
    if (Date.now() - lastFetchTime > 1000 * 60 * 5) { // 5 minutes
      fetchJobs();
    }
  }, 1000 * 60); // Check every minute

  return () => clearInterval(refreshInterval);
}, [lastFetchTime]);

const fetchJobs = async () => {
  try {
    setLoading(true);
    const response = await
```

```
fetch(`http://localhost:3000/api/jobs/user/${Cookies.get("id")}`);
  const data = await response.json();

  if (data?.jobs) {
    setJobs(data.jobs);
    setLastFetchTime(Date.now());
  } else {
    setError('No jobs found');
  }
} catch (error) {
  setError('Failed to fetch jobs');
  console.error('Error fetching jobs:', error);
} finally {
  setLoading(false);
}
};

if (loading && !jobs.length) {
  return (
    <div className="flex items-center justify-center min-h-screen">
      <Loader2 className="w-8 h-8 animate-spin text-blue-600" />
    </div>
  );
}

if (error && !jobs.length) {
  return (
    <div className="flex items-center justify-center min-h-screen text-red-600">
      {error}
    </div>
  );
}
```

```
return (  
  <div className="container mx-auto p-4 md:p-6">  
    <div className="flex justify-between items-center mb-8">  
      <h1 className="text-3xl font-bold">Available Positions</h1>  
      {loading && jobs.length > 0 && (  
        <div className="flex items-center gap-2 text-blue-600">  
          <Loader2 className="w-4 h-4 animate-spin" />  
          <span>Refreshing...</span>  
        </div>  
      )}  
    </div>  
  
    <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">  
      {jobs.map((job) => (  
        <div key={job.jobId} className="bg-white rounded-lg shadow-md  
hover:shadow-lg transition-shadow duration-300 overflow-hidden">  
          <div className="p-6">  
            <div className="flex items-start gap-4 mb-4">  
              <img  
                src={job.companyLogo || "/api/placeholder/48/48"}  
                alt={job.company}  
                className="w-12 h-12 rounded-lg object-cover"  
              />  
              <div>  
                <h2 className="text-xl font-semibold truncate">{job.title}</h2>  
                <div className="flex items-center gap-2 text-gray-600 mt-1">  
                  <Building2 size={16} />  
                  <span className="text-sm">{job.company}</span>  
                </div>  
              </div>  
            </div>  
          </div>  
        )  
      )  
    </div>  
  </div>  
)
```

```

<div
  className="text-gray-600 line-clamp-3 mb-4"
  dangerouslySetInnerHTML={{ __html: formatText(job.description) }}
/>

<div className="flex flex-wrap gap-4 mb-4">
  <div className="flex items-center gap-1 text-gray-600">
    <MapPin size={16} />
    <span className="text-sm">{job.location}</span>
  </div>
  <div className="flex items-center gap-1 text-gray-600">
    <Briefcase size={16} />
    <span className="text-sm">{job.employmentType}</span>
  </div>
</div>
</div>

<div className="border-t p-4 flex gap-3">
  <button
    onClick={() => setSelectedJob(job)}
    className="flex-1 flex items-center justify-center gap-2 px-4 py-2 border
rounded-md hover:bg-gray-50 transition-colors"
  >
    <Info size={16} />
    More Details
  </button>
  <a
    href={job.applyLink}
    target="_blank"
    rel="noopener noreferrer"
    className="flex-1 flex items-center justify-center gap-2 px-4 py-2 bg-blue-

```

```

600 text-white rounded-md hover:bg-blue-700 transition-colors"
    >
    <ExternalLink size={16} />
    Apply Now
  </a>
</div>
</div>
)}}
</div>

{selectedJob && (
  <div className="fixed inset-0 bg-black bg-opacity-50 flex items-center justify-
center p-4 z-50">
    <div className="bg-white rounded-lg max-w-3xl max-h-[80vh] overflow-y-auto
w-full">
      <div className="p-6">
        <div className="flex items-start justify-between mb-6">
          <div className="flex items-start gap-4">
            <img
              src={selectedJob.companyLogo || "/api/placeholder/48/48"}
              alt={selectedJob.company}
              className="w-12 h-12 rounded-lg"
            />
            <div>
              <h2 className="text-xl font-semibold">{selectedJob.title}</h2>
              <p className="text-gray-600">{selectedJob.company}</p>
            </div>
          </div>
          <div>
            <button
              onClick={() => setSelectedJob(null)}
              className="text-gray-500 hover:text-gray-700"
            >

```

```

        ×
    </button>
</div>

<div className="flex flex-wrap gap-4 mb-6">
    <div className="flex items-center gap-2 text-gray-600">
        <MapPin size={16} />
        <span>{selectedJob.location}</span>
    </div>
    <div className="flex items-center gap-2 text-gray-600">
        <Briefcase size={16} />
        <span>{selectedJob.employmentType}</span>
    </div>
    {selectedJob.postedAt && (
        <div className="flex items-center gap-2 text-gray-600">
            <Info size={16} />
            <span>Posted: {selectedJob.postedAt}</span>
        </div>
    )}
    {selectedJob.searchedSkill && (
        <div className="flex items-center gap-2 bg-blue-100 text-blue-800 px-2
py-1 rounded">
            <span className="text-sm">{selectedJob.searchedSkill}</span>
        </div>
    )}
</div>

<div className="prose max-w-none mb-6">
    <h3 className="text-lg font-semibold mb-2">Job Description</h3>
    <div
        className="whitespace-pre-wrap"
        dangerouslySetInnerHTML={{ __html:

```



```

formatText(selectedJob.description) }}
    />
  </div>

  <div className="flex justify-end">
    <a
      href={selectedJob.applyLink}
      target="_blank"
      rel="noopener noreferrer"
      className="flex items-center gap-2 px-6 py-2 bg-blue-600 text-white
rounded-md hover:bg-blue-700 transition-colors"
    >
      <ExternalLink size={16} />
      Apply Now
    </a>
  </div>
</div>
</div>
</div>
  )}
</div>
);
};

export default Jobs;

```

Narayana Career Connect


Available Positions Refreshing...


Python Developer (0 to 1 year)
 ConsoleCoder
 Join Us as a Python Developer! Experience: 0-1 Job Summary: We're looking for a skilled Python Developer with a passion for problem-solving and a talent for critical...
 India Full-time

More Details
Apply Now


Trainee Python Developer (0-1 Years) Harmo
 Harmony Data Integration Technologies Pvt. Ltd.
 Trainee Python Developer 0 1 Years Harmony Data Integration Technologies Pvt Ltd ResponsibilitiesSupport senior team members in designing and implementing data...
 Sahibzada Ajit Singh Nagar, Punjab Full-time

More Details
Apply Now


Python Software Engineer - SQL
 Jorim Technology Solutions Pvt Ltd
 Job Description Basic Qualification: • 0-1 year of experience in a similar role. • Strong foundational knowledge of Python and SQL. • Understanding of data modeling concepts and...
 Chennai, Tamil Nadu Full-time

More Details
Apply Now


Fresher Python Developer
 PrimefactorX
 Skills Python programming Problem solving Data structures Algorithms Object oriented programming Debugging Basic understanding of web development Version control Git...
 Chennai, Tamil Nadu Full-time

More Details
Apply Now


Junior Python Developer(Fresher)
 Alponet Digital Solutions Private Limited
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Fig: 4.4.2. Explore Jobs

4.3. SUMMARY

The development and implementation of **Narayana Career Connect**, an innovative career development and job placement web application, mark a significant advancement in bridging the gap between job seekers and mentors. The platform is designed to provide an efficient, user-friendly experience for individuals seeking employment opportunities, skill enhancement, and professional mentorship.

Key functionalities include **intelligent job matching**, which utilizes user-inputted skills and preferences to recommend the most relevant job opportunities, streamlining the job search process. The **mentorship and networking module** enables users to connect with industry professionals, fostering career growth through guidance and insights.

Additionally, the platform integrates **career development resources**, including courses, industry insights, and success stories, empowering users with the knowledge and motivation to advance in their careers.

Narayana Career Connect's initial deployment has demonstrated notable improvements in **user engagement, job placement rates, and mentorship participation**. By providing a structured and accessible career development ecosystem, the platform supports individuals in achieving their professional goals, enhancing employability, and fostering a culture of continuous learning and growth.

System Implementation Modules

- **User Interface**
- **User Management**
- **Job Search**
- **Mentorship & Networking**
- **Messages**
- **Career Development Resources**

5.SYSTEM TESTING

5.1.. INTRODUCTION

System testing is a crucial phase within the **Narayana Career Connect** web application's development lifecycle, dedicated to verifying that the entire system operates seamlessly and adheres to predefined requirements. Through a series of meticulously designed tests, the aim is to identify and address any defects or issues that may arise during the development process. The primary goal of system testing in the **Narayana Career Connect** project is to guarantee that the system functions reliably, securely, and efficiently, thereby providing users with a flawless experience.

In the **Narayana Career Connect** application, system testing is instrumental in validating the functionality, integration, and performance of various components such as **job search, mentorship connections, career development resources, and application tracking**. By conducting comprehensive testing, we ensure that the system meets end-to-end specifications and operates effectively within a real-world web-based environment.

System testing in the **Narayana Career Connect** project encompasses a range of tests designed to exercise the complete system, ensuring its robustness, accuracy, and usability. This rigorous testing approach enhances the quality and reliability of the **job placement and mentorship process**, reinforcing the application's capability to deliver a seamless experience for users.

5.2. TESTING PROCESS

The testing process for the **Narayana Career Connect** web application follows a structured approach to ensure comprehensive coverage and reliability. This involves **planning, designing test cases, executing tests, and evaluating results** to ensure the system functions correctly and meets user expectations.

The methodologies employed include:

- **Unit Testing** – Verifying individual components like job search filters, profile management, and messaging features.
- **Integration Testing** – Ensuring smooth interaction between different modules such as job applications, mentorship features, and user dashboards.
- **Black Box Testing** – Assessing the system's functionality from an end-user perspective without focusing on internal code structures.
- **White Box Testing** – Analyzing internal workflows, including database interactions and API functionalities.

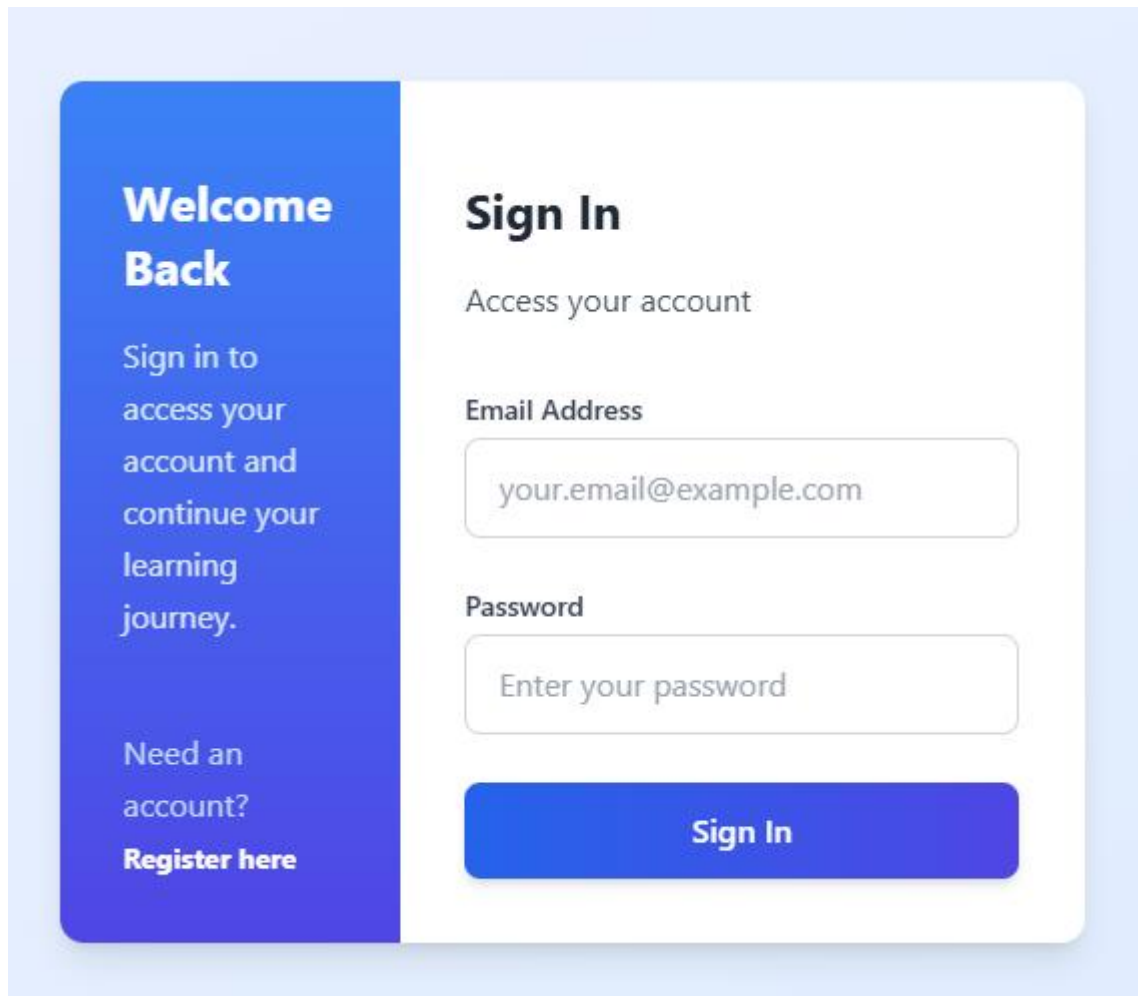
5.3 UNIT TESTING

Unit testing is the process of testing the smallest functional unit of code. It helps ensure code quality and is an integral part of software development. As a best practice, software is developed as small, functional units, with each unit having a corresponding unit test to verify its correctness.

The following are some of the test cases executed during the **Narayana Career Connect** testing process:

TC01: Verify user login functionality

- **Description:** The system should allow registered users to log in successfully.
- **Pre-conditions:** User account exists, valid login credentials available.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Enter a valid email and password.
 3. Click on the "Login" button.
- **Expected Result:** User is successfully logged in and redirected to the dashboard.
- **Actual Result:** User is successfully logged in and redirected to the dashboard.



The image shows a login interface for 'Narayana Career Connect'. It features a blue sidebar on the left with the text 'Welcome Back' and 'Sign in to access your account and continue your learning journey.' Below this is a link 'Need an account? Register here'. The main white area on the right is titled 'Sign In' and contains the subtext 'Access your account'. It has two input fields: 'Email Address' with the placeholder 'your.email@example.com' and 'Password' with the placeholder 'Enter your password'. A blue 'Sign In' button is at the bottom of the form.

Welcome Back

Sign in to access your account and continue your learning journey.

Need an account? [Register here](#)

Sign In

Access your account

Email Address

your.email@example.com

Password

Enter your password

Sign In

Fig. 5.1.3 Login

- **Status:** Pass

TC02: Test Dashboard Page Functionality

- **Description:** The dashboard page should load successfully and display relevant user information.
- **Pre-conditions:** User is logged in with an active account.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Log in with valid credentials.
 3. Check if job recommendations, mentorship connections, and notifications are displayed.
- **Expected Result:** The dashboard page loads successfully, displaying personalized job recommendations, mentorship options, and career-related updates.
- **Actual Result:** The dashboard page loads successfully, displaying personalized job recommendations, mentorship options, and career-related updates.

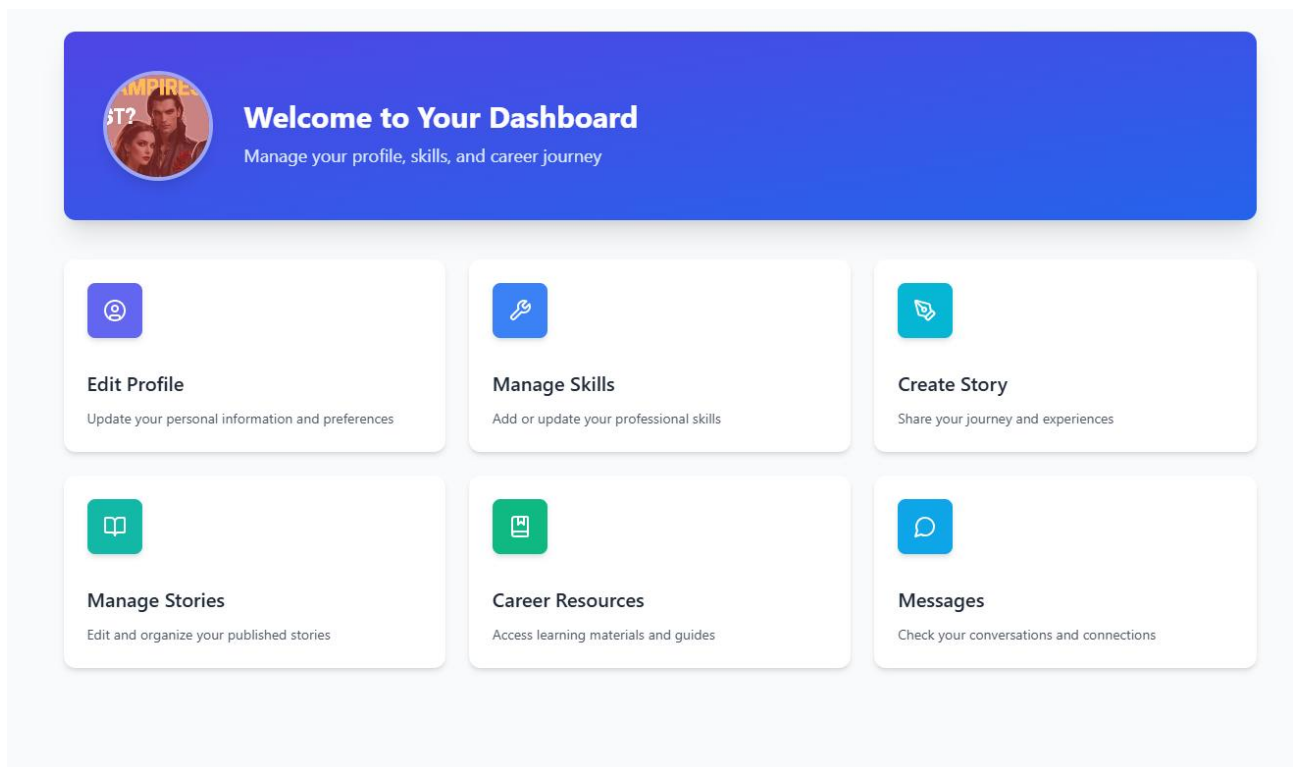


Fig. 5.3.2. Dashboard

- **Status:** Pass

TC03: Verify Add Skills Functionality

- **Description:** The system should allow users to add their skills successfully.
- **Pre-conditions:** User is logged in and navigates to the profile or skills section.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate to the "Profile" or "Skills" section.
 3. Click on the "Add Skills" button.
 4. Enter or select relevant skills from the list.
 5. Save the changes.
- **Expected Result:** The selected skills are added and displayed in the user's profile.
- **Actual Result:** The selected skills are added and displayed in the user's profile.

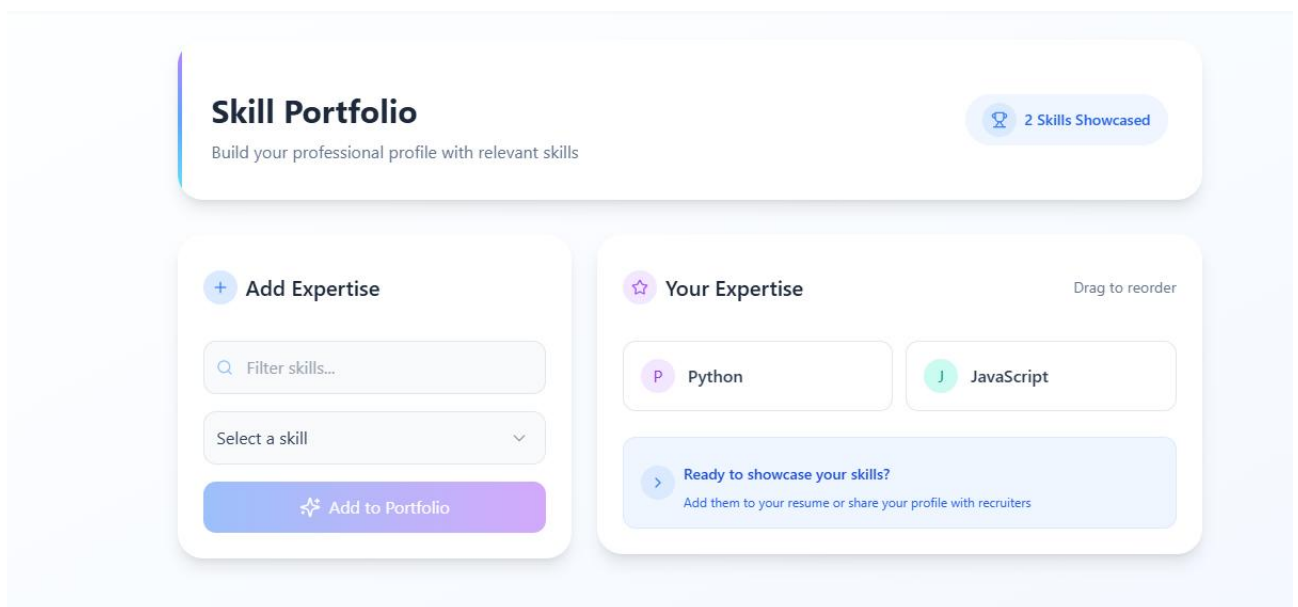


Fig. 5.3.3 Manage Skills

- **Status:** Pass

TC04: Messages in Connections

- **Description:** Appropriate error messages should be displayed when a user attempts to send an invalid message to a connection.
- **Pre-conditions:** User is logged in and has at least one connection.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate to the "Connections" or "Messages" section.
 3. Select a connection and open the chat window.
 4. Click on the "Send" button.
- **Expected Result:** A relevant error message is displayed, preventing the invalid message from being sent.
- **Actual Result:** A relevant error message is displayed, preventing the invalid message from being sent.

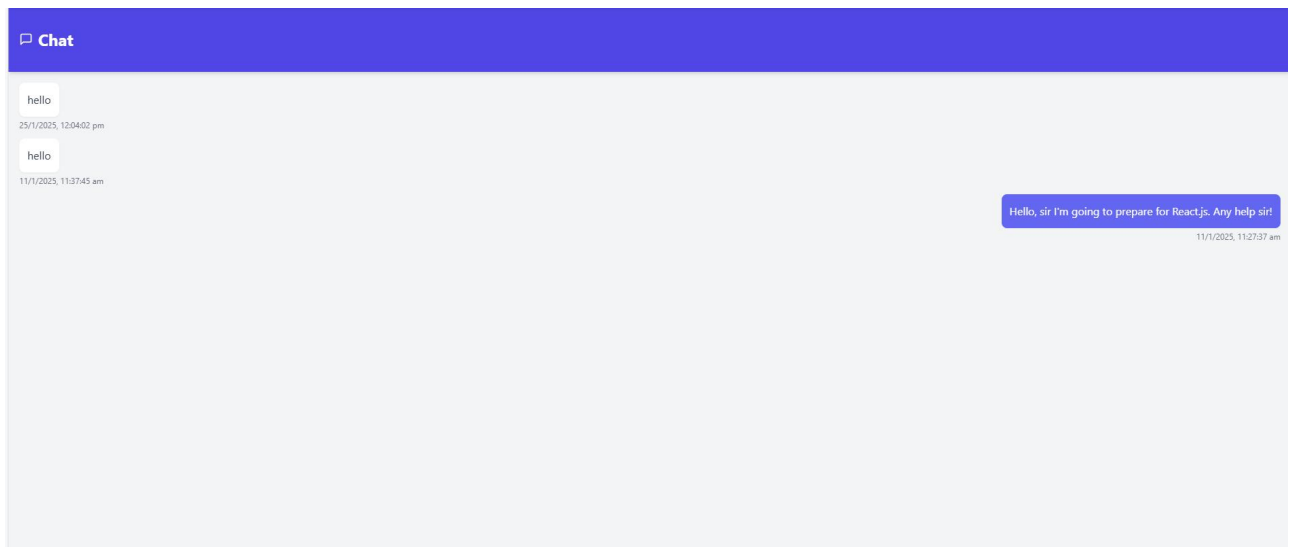


Fig.5.3.4. Chat System

TC05: Test Explore Connections Page Functionality

- **Description:** The "Explore Connections" page should load successfully and display relevant mentor or peer connections.
- **Pre-conditions:** User is logged in and has access to the "Explore Connections" feature.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate to the **"Explore Connections"** page.
 3. Verify that suggested mentors, industry professionals, or peers are displayed.
 4. Attempt to send a connection request.
- **Expected Result:** The "Explore Connections" page loads successfully, displaying suggested connections, and the user can send connection requests.
- **Actual Result:** The "Explore Connections" page loads successfully, displaying suggested connections, and the user can send connection requests.
- **Status:** Pass

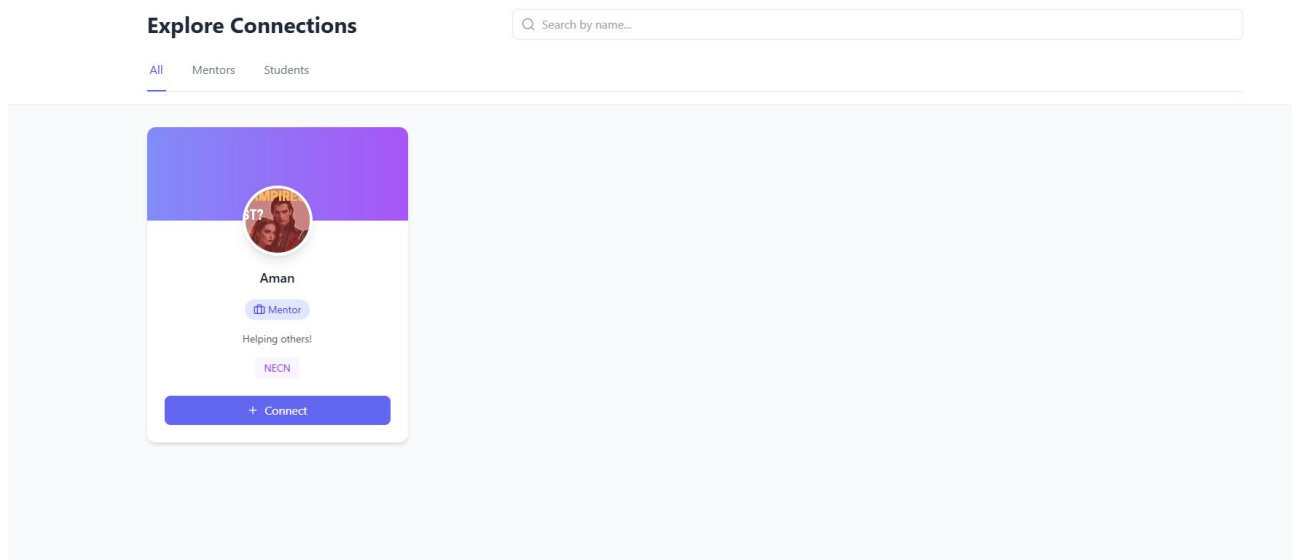


Fig. 5.3.5 Explore Connections

- **Status:** Pass

5.4 INTEGRATED TESTING

Integration testing is the second level of the software testing process after the unit testing process. This testing phase is crucial to ensure seamless interactions among various units/components, their functionalities, and how well they can operate as a single entity. Developing comprehensive integration test cases is essential. These cases cover various scenarios, including data flow, error handling, performance, and security. By considering boundary conditions, edge cases, and negative scenarios, testers ensure the robustness and reliability of the integrated system. The following are test cases

TC01: Verify Internet Connectivity Handling

- **Description:** The **Narayana Career Connect** web application should access the internet and perform expected operations without issues.
- **Pre-conditions:** Web application is deployed, and an active internet connection is available.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate through different sections (Dashboard, Job Search, Mentorship).
 3. Monitor the system's response when the internet connection is stable and when disconnected.
- **Expected Result:** The system functions properly with an active internet connection and displays an appropriate error message when offline.
- **Actual Result:** The system functions properly with an active internet connection and displays an appropriate error message when offline.
- **Status:** Pass

TC02: Test User Login Functionality

- **Description:** A registered user should be able to log in successfully.
- **Pre-conditions:** Web application is accessible, and the user has valid login credentials.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate to the **Login** page.
 3. Enter valid credentials (email/username and password).
 4. Click on the **Login** button.
- **Expected Result:** The user is successfully authenticated and redirected to the **Dashboard**.
- **Actual Result:** The user is successfully authenticated and redirected to the **Dashboard**.
- **Status:** Pass

TC03: Test System Exploring for Job & Mentorship Resources

- **Description:** The **Narayana Career Connect** web application should post notifications for job updates, mentorship messages, or application status changes.
- **Pre-conditions:** User is logged in and has enabled notifications.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Communicate through messages and resources
- **Expected Result:** A notification is posted in the system's notification area.
- **Actual Result:** A notification is posted in the system's notification area.
- **Status:** Pass

TC04: Verify Job-Mentor Matching System

- **Description:** The application should recommend jobs or mentors based on the user's skills and preferences.
- **Pre-conditions:** User has a completed profile with skills and preferences set.
- **Test Steps:**
 1. Open the **Narayana Career Connect** web application.
 2. Navigate to the **Job Search** or **Mentorship** section.
 3. Search for jobs or mentors based on skills and industry preference.
 4. Display matching job opportunities or mentor connections.
- **Expected Result:** The system displays job listings or mentors relevant to the user's skills and preferences.
- **Actual Result:** The system displays job listings or mentors relevant to the user's skills and preferences.
- **Status:** Pass

5.5 BLACKBOX TESTING

Black box testing involves testing the functionality of the application without knowing the internal code structure. This typically includes integration tests and tests based on user interactions. Black box testing is a software testing technique where the internal structure, design, or implementation of the system being tested is not known to the tester. Instead, the tester focuses solely on the external behavior of the software, treating it as a black box where inputs are provided, and outputs are observed. Various techniques, such as equivalence partitioning, boundary value analysis, and decision table testing, are employed to design comprehensive test cases covering different usage scenarios. The following are testcases

TC01: Verify Internet Connectivity Handling

- **Description:** The Narayana Career Connect web application should access the internet and perform expected operations.
- **Components Integrated:** Network services, data synchronization.
- **Expected Result:** Successful internet access and operations.

TC02: Test System Notifications for Job & Mentorship Updates

- **Description:** The system should post notifications for job updates, mentorship messages, or application status changes.
- **Components Integrated:** Notification services, event triggers.
- **Expected Result:** Notification is posted.

TC03: Test Dashboard Navigation

- **Description:** The application should allow smooth navigation across different pages without lag or crashes.
- **Components Integrated:** Navigation, UI components.
- **Expected Result:** Smooth navigation.

TC04: Test Job & Mentorship Recommendations Notifications

- **Description:** Users should receive relevant job alerts and mentorship recommendations.
- **Components Integrated:** Notification services, user preferences, job-matching system.
- **Expected Result:** Notifications are sent based on user preferences.

TC05: Verify Data Synchronization

- **Description:** User profile updates, job applications, and messages should sync correctly with the server.
- **Components Integrated:** Data synchronization services.
- **Expected Result:** Data is synced successfully.

5.6 WHITEBOX TESTING

White box testing involves testing the internal structures or workings of an application. This typically includes unit tests and tests that require knowledge of the code structure. White box testing, also known as structural or glass box testing, scrutinizes the internal structure and implementation details of a software system. Unlike black box testing, which focuses solely on external behaviour, white box testing examines code paths, logic flows, and data structures within the software. Testers, armed with knowledge of the source code, design test cases to ensure thorough coverage of statements, branches, paths, and loops. Test cases are as follows

TC01: Verify User Skills Input Request

- **Description:** The system should prompt users to enter their skills upon profile setup or job search.
- **Components Tested:** Skill input form, user profile module.
- **Expected Result:** A prompt appears requesting the user to input their skills.

TC02: Test Skills-Based Job & Mentor Recommendations

- **Description:** The system should display job and mentor suggestions based on user-entered skills.
- **Components Tested:** Recommendation engine, database query, UI display.
- **Expected Result:** Relevant jobs and mentors are displayed based on the user's skills.

TC03: Verify App Launch

- **Description:** The **Narayana Career Connect** web application should launch without errors and display the dashboard.
- **Components Tested:** App initialization, authentication mechanism.
- **Expected Result:** The dashboard is displayed without crashes.

TC04: Verify UI Elements Visibility on Dashboard

- **Description:** All essential UI elements, such as job postings, mentor suggestions, and navigation, should be visible and properly aligned.
- **Components Tested:** UI components, front-end design, CSS layout.
- **Expected Result:** UI elements are visible and aligned correctly.

TC05: Test Skill Addition & Form Submission

- **Description:** Users should be able to add their skills and submit the form successfully.
- **Components Tested:** Skill input form, backend validation, database updates.
- **Expected Result:** Skills are successfully saved, and confirmation feedback is provided.

TC06: Verify Error Messages for Invalid Skill Input

- **Description:** The system should display relevant error messages if a user enters invalid or empty skills.
- **Components Tested:** Form validation, error-handling mechanism.
- **Expected Result:** The system displays correct error messages for invalid skill entries.

5.7 SUMMARY

In developing the **Narayana Career Connect** web application, a comprehensive testing strategy is crucial to ensure functionality, reliability, and a seamless user experience. The testing approach includes **unit testing, integration testing, black-box testing, and white-box testing** to validate different aspects of the platform.

Unit testing focuses on individual components such as user authentication, skill addition, job recommendations, and mentorship connections. This ensures that essential functionalities, including user login, skill input, profile updates, and job applications, operate correctly in isolation.

Integration testing verifies interactions between various modules, ensuring smooth data flow and system performance. This includes validating job and mentorship recommendations based on user skills, real-time data synchronization, and seamless communication between users and mentors.

Black-box testing evaluates the system from an end-user perspective, ensuring intuitive navigation and UI functionality. Key aspects tested include dashboard responsiveness, job search efficiency, mentorship suggestions, and proper handling of error messages for invalid input. This guarantees a smooth user experience across different scenarios.

White-box testing examines the internal logic, algorithms, and system architecture to ensure efficiency and reliability. This includes validating background processes, optimizing database queries for job-matching efficiency, and ensuring the platform performs well under varying loads.

6.RESULTS AND DISCUSSION

The **Narayana Career Connect** platform has significantly enhanced the efficiency of career networking, job searching, and mentorship connections. It has undergone rigorous testing and validation, proving its effectiveness in managing skill-based job recommendations, mentor-mentee interactions, and professional networking. Users have reported substantial time savings and a reduction in manual job searches due to its structured recommendations. Stakeholder feedback highlights its intuitive user interface and seamless workflows, fostering better collaboration between job seekers, recruiters, and mentors. This platform marks a major advancement in modernizing career development and provides opportunities for continuous improvements to align with evolving industry trends.

The integration of **Narayana Career Connect** into the professional networking ecosystem has streamlined job discovery and mentorship access. Additionally, it provides valuable insights and analytics to enhance user experience and optimize career strategies. By tracking key metrics such as skill demand, job application trends, and mentorship engagement, career advisors and recruiters gain actionable intelligence to refine hiring and mentoring initiatives. The platform stands as a crucial step forward in career growth facilitation, offering automated job-matching, targeted notifications, and data-driven career insights.

The primary outcome of the project is to establish a structured and efficient process for connecting job seekers with opportunities and mentors. The system enables users to receive personalized job recommendations and mentorship suggestions based on their skills and interests, ensuring that professional development is both relevant and impactful. Additionally, the platform prioritizes real-time updates and notifications, keeping users informed about job openings, mentorship opportunities, and career events, enhancing engagement and minimizing missed opportunities.

A key feature of **Narayana Career Connect** is its structured skill-matching system, which recommends jobs and mentors based on users' skill sets and career goals. This functionality is crucial in providing targeted career guidance and ensuring meaningful professional connections.

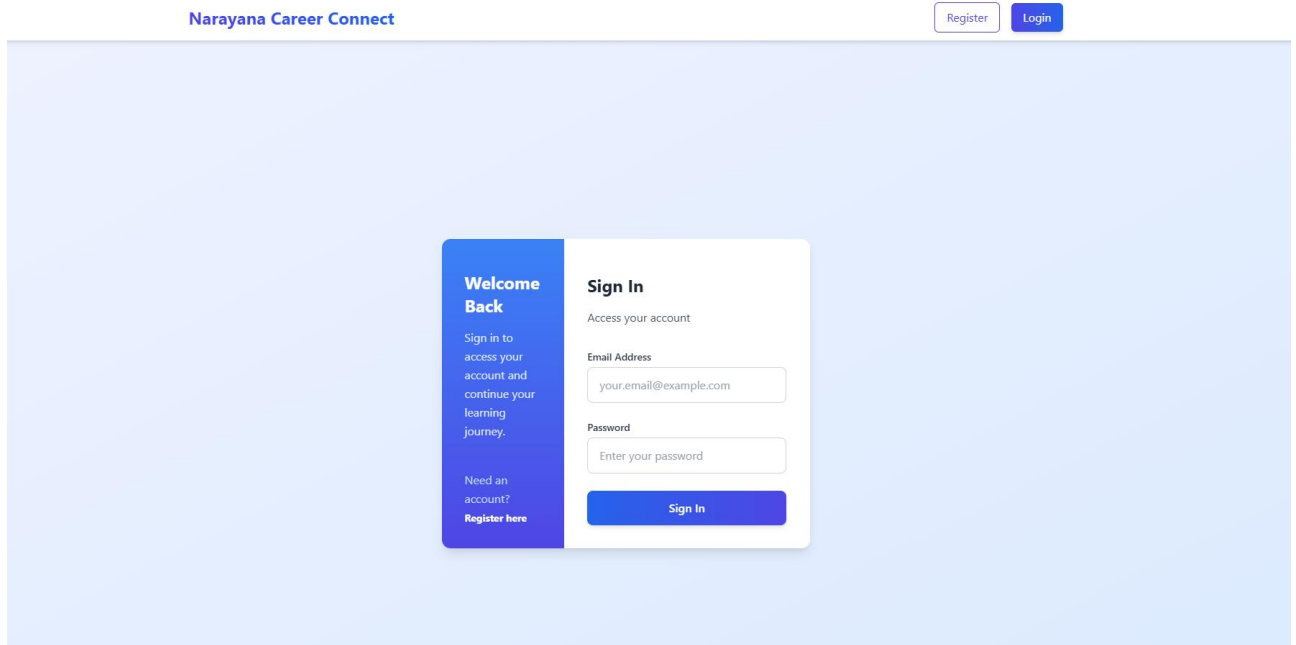


Fig. 6.1. Login Screen

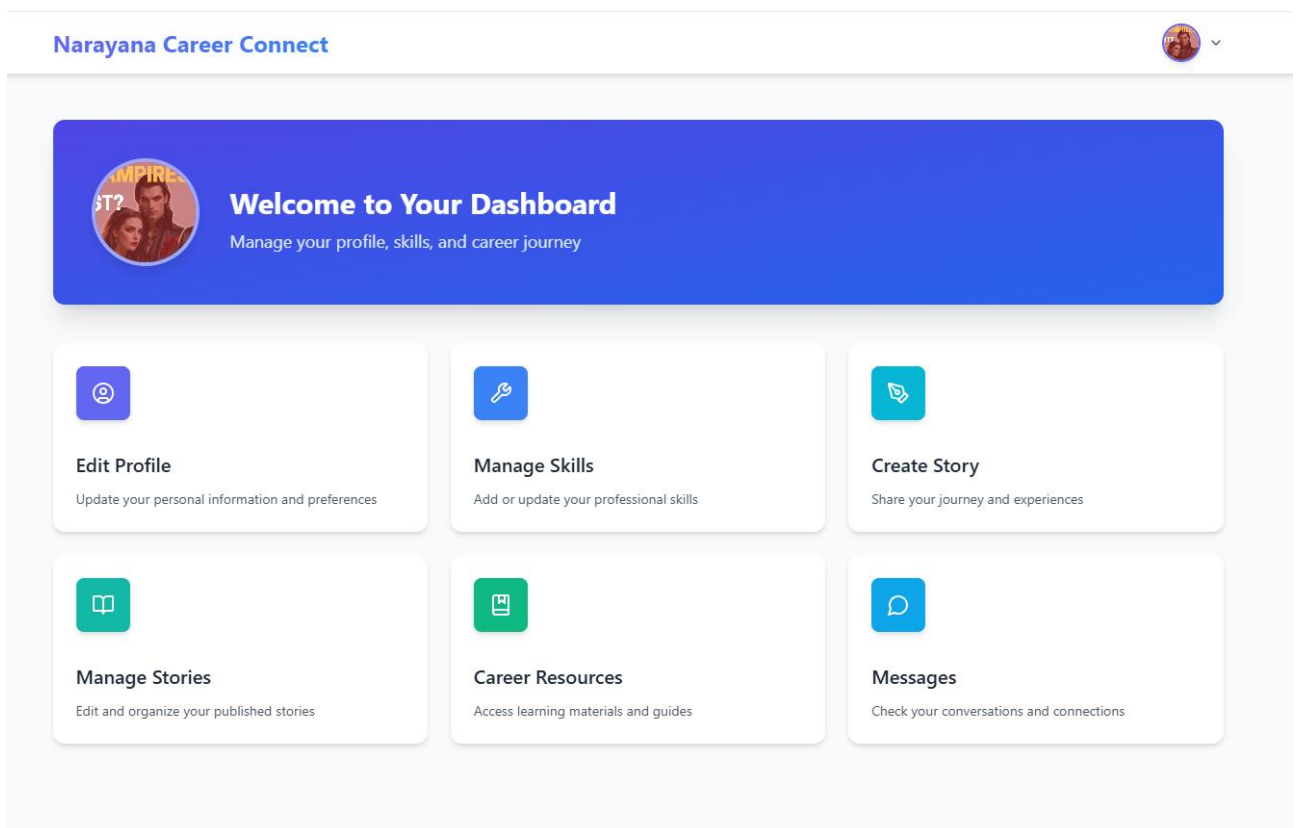


Fig. 6.2. Dashboard Screen

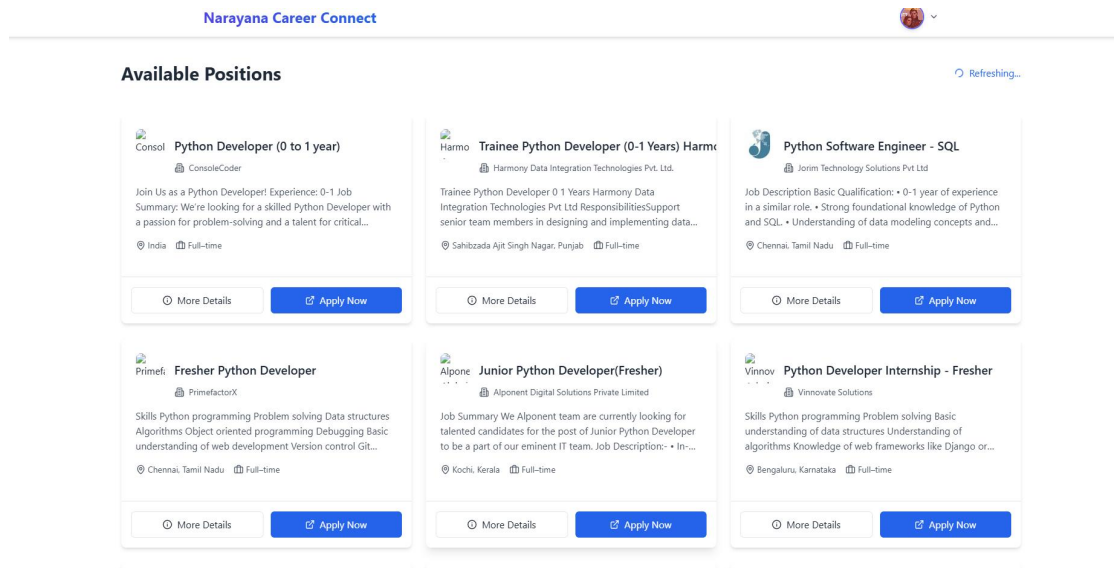


Fig. 6.3. Explore Jobs Page

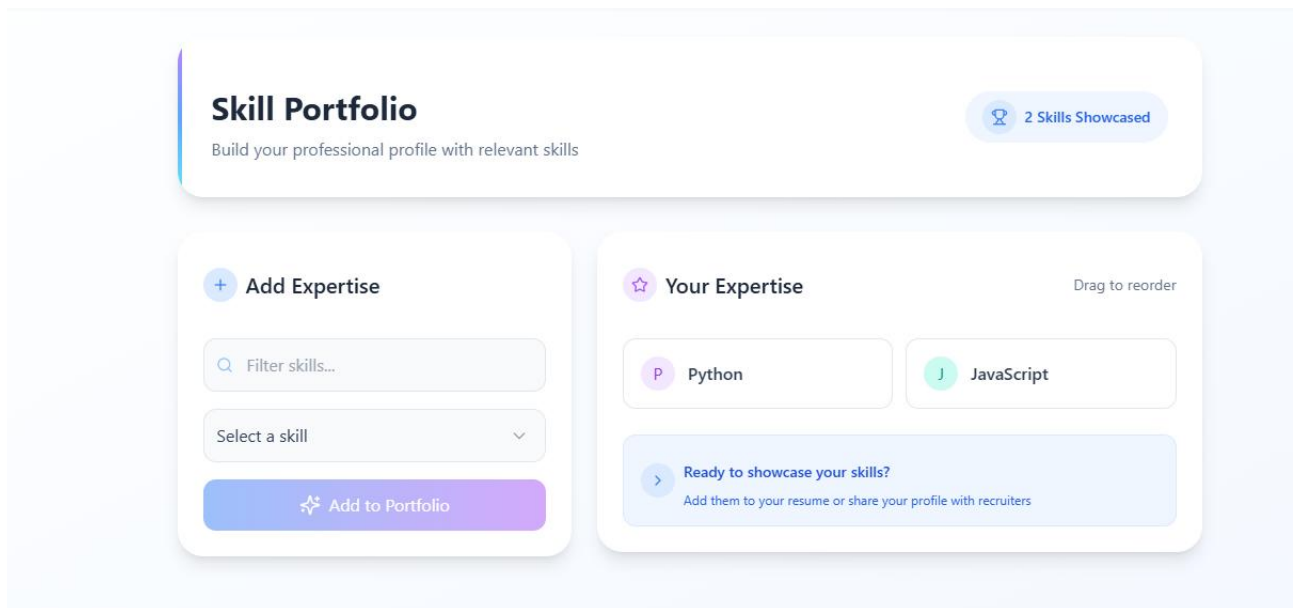


Fig. 6.4 Manage Skills Page

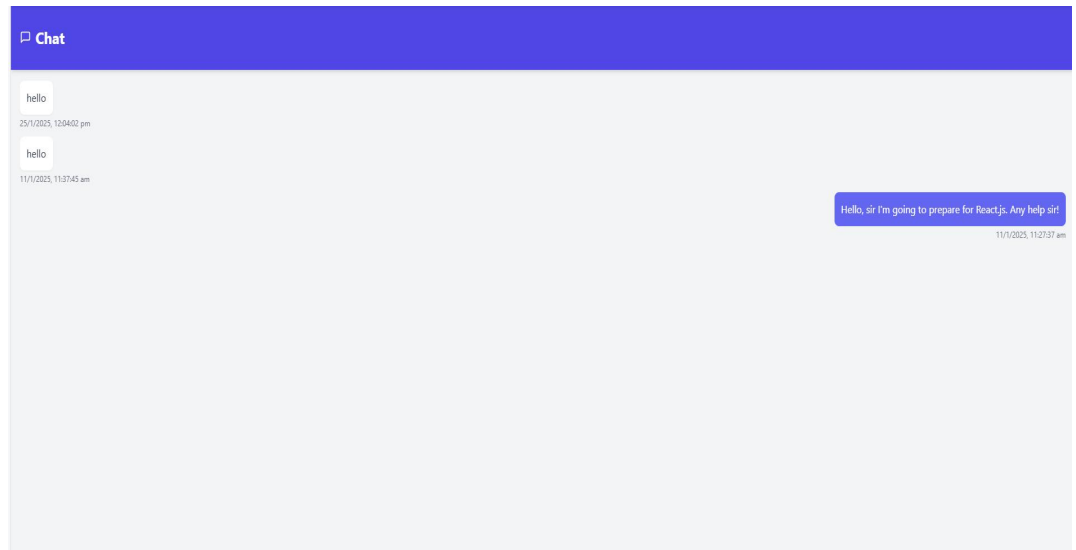


Fig. 6.5. Chat System

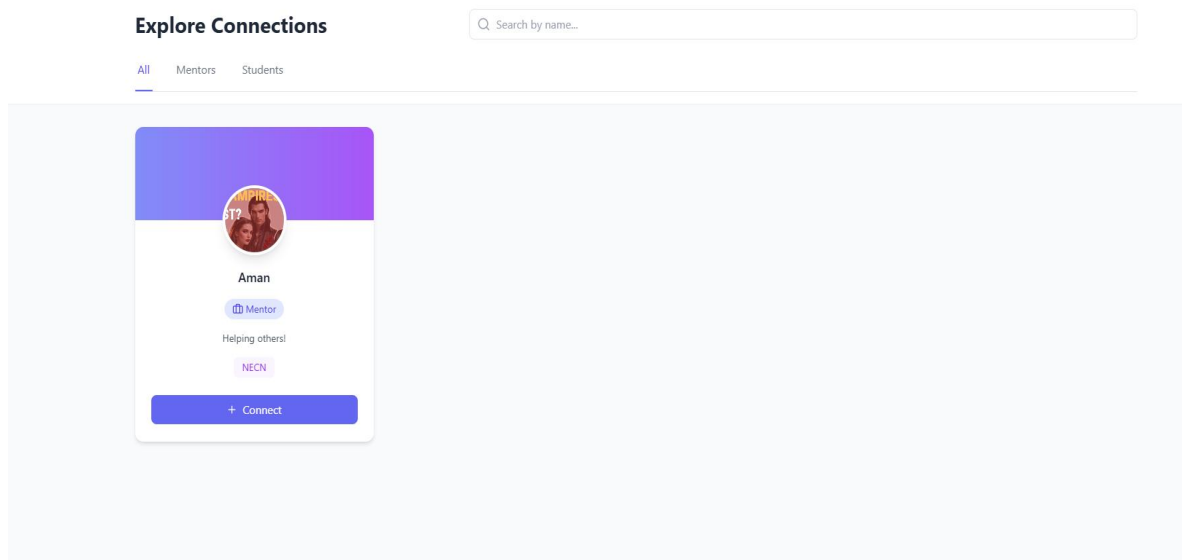


Fig 6.6. Explore Connections

7. CONCLUSION AND FUTUE SCOPE

7.1 CONCLUSION

The web-based **Career Guidance and Job Assistance Application** represents a transformative advancement in career development and employment accessibility. By integrating automated job-matching, mentorship programs, and skill enhancement resources, the platform streamlines career planning, empowering job seekers with personalized guidance and real-time opportunities. This automation reduces manual effort for both job seekers and employers, making the job search and hiring process more efficient and precise.

Additionally, the inclusion of **community engagement features** fosters an interactive environment where professionals, mentors, and recruiters collaborate, share insights, and support career growth. Through discussion forums, career webinars, and expert advice, users gain access to valuable industry knowledge and networking opportunities. This interactive approach enhances user engagement, ensuring a dynamic and supportive career ecosystem.

Through rigorous testing and user feedback, the system has demonstrated its **reliability, scalability, and effectiveness** in facilitating career advancements. Users have reported a seamless experience in navigating job listings, receiving mentorship, and enhancing their professional skills. The application's intuitive design and data-driven approach further strengthen its impact on career success.

In conclusion, the **Career Guidance and Job Assistance Application** serves as a **comprehensive and innovative platform** that bridges the gap between job seekers, mentors, and employers. By **leveraging automation, real-time insights, and community collaboration**, it significantly enhances career development processes, contributing to a **more efficient, informed, and empowered workforce** in the ever-evolving job market.

7.2 FUTURE SCOPE

The Career Guidance and Job Assistance Application has a promising future with several potential improvements that can make it even more effective and user-friendly. One major enhancement could be expanding career guidance resources specific to different industries. By offering detailed career roadmaps, job market insights, and skill development recommendations for fields like IT and engineering, users can receive more personalized support in their career journey.

Another valuable addition would be real-time job and internship updates through stronger partnerships with recruiters and companies. This would ensure users have access to the latest job openings, internship opportunities, and campus recruitment drives, helping them stay ahead in their job search.

Introducing interactive mentorship programs could also make a significant difference. Through career counseling sessions, webinars, and live Q&A forums, users can connect directly with experienced professionals and industry experts for advice on career planning, resume building, and interview preparation.

A built-in skill development and certification tracking feature could further enhance the platform's effectiveness. By allowing users to enroll in courses, track their progress, and showcase certifications, this feature would help improve employability and ensure they meet industry standards.

Finally, expanding community engagement through discussion forums, success stories, and networking groups would create a supportive space for users to share experiences, ask questions, and stay motivated throughout their career journey.

With these future improvements, the Career Guidance and Job Assistance Application will continue to evolve as a powerful tool that connects users with the right opportunities, resources, and guidance, ultimately helping them build successful careers.

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